

INFRASTRUCTURE REFERENCE — CONFIDENTIAL

Connector

Platform Architecture Reference



Connector Platform Team

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Chapter 1

Introduction

1.1 What Is Connector?

Connector is an **infrastructure-grade agentic runtime**: a platform that executes autonomous AI agents the way an operating system executes processes — with governed memory, typed communication channels, resource budgets, audit trails, and cryptographic proof of every state transition.

Most AI agent frameworks are *libraries*: they give developers helpful abstractions for calling LLMs and chaining tool calls, but they leave governance, auditability, recovery, and cost control entirely to the application layer. In production environments — healthcare, finance, legal, enterprise — this is insufficient. When an agent makes a decision, operators need to know *what* it decided, *why*, *what data it accessed*, *how much it cost*, and *whether a human can verify it* after the fact.

Connector solves this at the infrastructure level, not the application level.

Core Value Proposition

Every agent decision on Connector is:

- **Recorded** — written to a signed, content-addressed memory packet
- **Auditable** — every access, tool call, and state transition is logged
- **Traceable** — a full decision chain can be reconstructed at any time
- **Provable** — cryptographic receipts allow independent verification
- **Governed** — budgets, predicates, and compliance rules are enforced by the kernel

1.2 The Problem Space

Running AI agents in production creates a set of hard infrastructure problems that cannot be solved by wrapping LLM APIs:

1. **State fragility** — Agent state lives in memory and is lost on crash. Connector persists all state as immutable memory packets with lineage tracking.
2. **No audit trail** — Log lines are not evidence. Connector writes cryptographically signed receipts for every operation.
3. **Unbounded cost** — LLM token spend is invisible until the bill arrives. Connector enforces per-session and per-agent token budgets in real time.
4. **No governance** — Compliance requirements (HIPAA, SOC2, GDPR) cannot be bolted on after the fact. Connector treats predicates and compliance rules as first-class kernel primitives.

5. **Protocol chaos** — Agents need to communicate with each other, with external tools, and with humans through different transports. Connector provides a unified protocol stack (CNP) with bridges to A2A and MCP.

1.3 Architecture Overview

Connector is structured as seven discrete layers, each building on the one below. The architecture is intentionally kernel-inspired: lower layers provide stable primitives; upper layers compose those primitives into product surfaces.

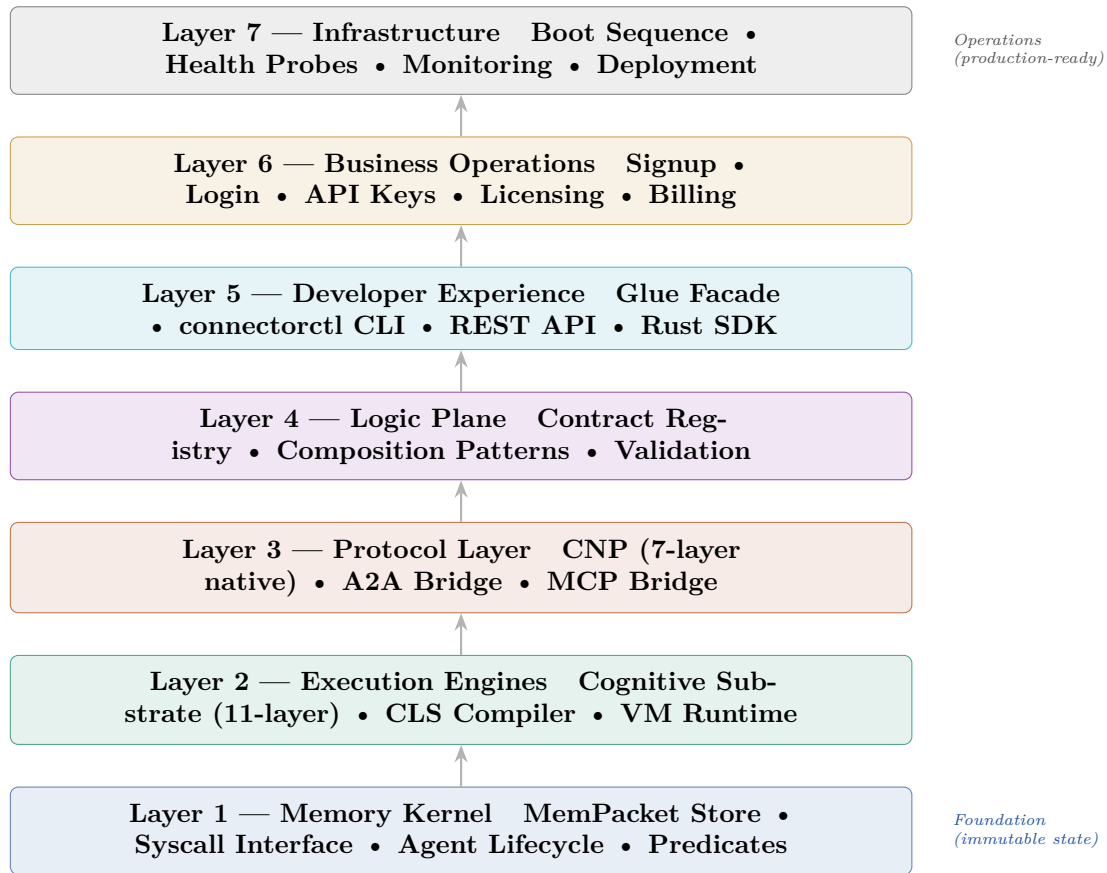


Figure 1.1: Connector 7-layer architecture — each layer provides stable primitives to the layer above

The critical insight is **directionality**: requests flow *down* through the stack, and evidence flows *up*. A developer calls a Glue verb; the execution engine interprets it; the kernel enforces governance and writes a signed receipt; that receipt bubbles back up through the stack as proof.

1.4 Design Principles

Seven principles govern every architectural decision in Connector:

Principle	What it means in practice
Memory-First	All state is a MemPacket. Computation is memory transformation. Nothing is ephemeral unless explicitly marked as such.
Content Addressing	Every artifact is identified by its content hash (CID). Identical content is deduplicated; any tampering changes the CID.
Governance by Default	Predicates, budgets, and audit requirements are enforced by the kernel before any agent operation executes — not after.
Recovery by Default	Every layer checkpoints state. Node crash, budget violation, or network partition never causes unrecoverable data loss.
Evidence Everywhere	Every state transition produces a cryptographic receipt. Auditors can verify the full history of any agent at any time.
Protocol Agnostic	CNP is native, but A2A and MCP bridges mean Connector agents interoperate with the broader AI ecosystem out of the box.
Operator Ergonomics	The Glue facade and <code>connectorctl</code> CLI expose complex primitives through a simple, human-readable grammar.

Table 1.1: Connector’s seven design principles

1.5 Deployment Models

Connector supports four deployment topologies:

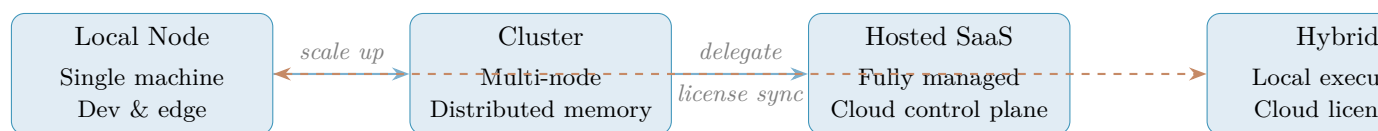


Figure 1.2: Connector deployment topology options

- **Local Node** — A single binary (`connector-platform`) runs the entire stack on one machine. Used for development, testing, and edge deployments.
- **Cluster** — Multiple nodes share a distributed memory store with consensus for high availability and horizontal scale.
- **Hosted SaaS** — The Connector team operates the control plane; customers access it via the customer portal and API keys.
- **Hybrid** — Execution stays on-premise (data sovereignty), while licensing, billing, and the admin dashboard run on the hosted control plane.

1.6 Key Terminology

Glossary	
MemPacket	Immutable, content-addressed memory object. The atomic unit of all state in Connector. Every MemPacket has a CID, typed content, metadata (timestamps, lineage), and an optional Ed25519 signature.
CID	Content Identifier — <code>cls1-sha256-<64-char-hex></code> . Computed as the SHA-256 hash of the canonical JSON representation of the packet's content. Any change to content produces a different CID.
Agent	Autonomous execution unit with a DID (decentralised identifier), memory context, capability grants, and a lifecycle state machine (Registered → Active → Paused → Terminated).
Session	Scoped execution context. Carries a resource budget (token limit, memory quota, wall-clock limit), a set of governance predicates, and a checkpoint chain.
Contract	Compiled, signed executable logic produced by the CLS compiler. Contains an interface specification, a state machine, governance rules, and a resource envelope.
CNP	Connector Native Protocol — a 7-layer typed communication stack for agent-to-agent messaging with capability gating, encryption, and delivery guarantees.
CLS	Connector Logic Standard — the contract language, 6-pass compiler, and runtime VM that transform agent intent declarations into governed, verifiable execution.
Glue	The developer-facing surface of Connector. Provides action-first verbs (start , stop , list , trace , prove) over CLI, REST API, and Rust library interfaces.
SOE	Surface Output Engine — the layer that produces signed, CID-anchored output documents (reports, receipts, defense packages) for auditors and operators.

1.7 How to Read This Document

This document progresses *bottom-up* through the architecture stack, mirroring how the platform is built. Readers who want to understand how an operator deploys and operates Connector should read Chapters 6–7 first. Readers building agents or integrations should focus on Chapters 2–5.

Ch.	Layer	Covers
2	Memory Kernel	MemPacket anatomy, CID addressing, syscall categories, predicate enforcement, agent life-cycle, cryptographic receipts
3	Execution Engines	11-layer Cognitive Substrate, CLS 6-pass compiler, contract runtime, resource budgets
4	Protocol Layers	CNP 7-layer stack, A2A and MCP bridges, message types, capability attenuation, delivery guarantees
5	Developer UX	Glue verb grammar, connectorctl CLI, REST API surface, configuration hierarchy, error model
6	Business Ops	Two-server architecture, signup/login flows, API key issuance, license tiers, Stripe billing integration
7	Node Operations	12-stage boot sequence, health probes, agent deployment manifest, monitoring, disaster recovery

Table 1.2: Chapter guide by audience and layer

Chapter 2

Memory Kernel

2.1 The Foundation Layer

The Memory Kernel is the bedrock of the entire Connector platform. Every other layer — execution engines, protocol stacks, developer tooling — sits on top of it. Its job is deceptively simple: **ensure that all agent state is persistent, immutable, addressable by content, and cryptographically verifiable**. Nothing happens in Connector that is not mediated by the kernel.

The best analogy is an operating system kernel combined with a content-addressed filesystem. An OS kernel mediates every system call a process makes; Connector's Memory Kernel mediates every operation an agent performs. An OS filesystem tracks file paths and permissions; the Memory Kernel tracks content hashes and cryptographic lineage.

Why content addressing?

Traditional databases identify records by primary key — an arbitrary number that tells you nothing about the data. Content addressing identifies records by their *content hash*. If you have the CID, you can verify the content is unchanged without trusting any server. Identical content produces identical CIDs, giving automatic deduplication. This is how Connector provides tamper-evident audit trails without a separate audit database.

2.2 The MemPacket

A **MemPacket** is the atomic unit of all state in Connector. Every piece of information — an agent's reasoning output, a tool call result, a compliance record, a session checkpoint — is stored as a MemPacket. There is no other storage primitive.

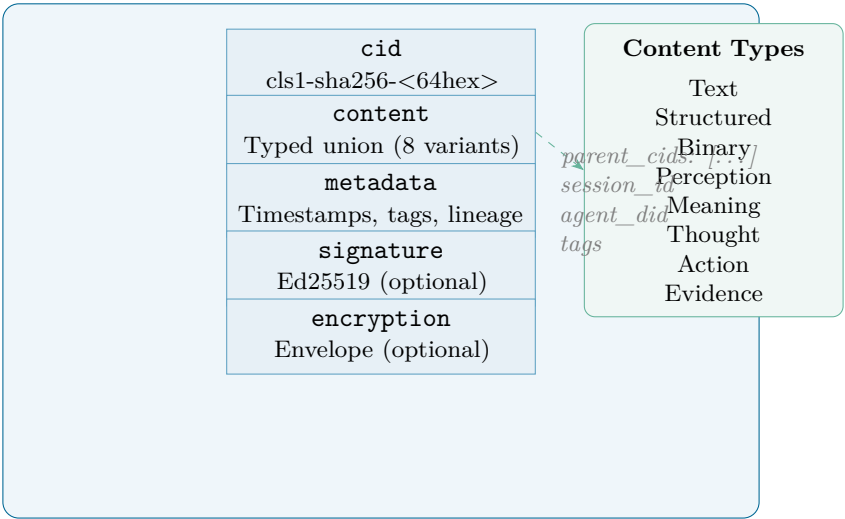


Figure 2.1: MemPacket anatomy — the atomic state unit of Connector

2.2.1 Content Types

The **content** field is a typed union with eight variants, covering the full range of information an agent might produce:

Type	Use case	Typical producer
Text	Raw text, prompts, responses	LLM gateway
Structured	JSON data, query results	Tool calls, API responses
Binary	Files, images, embeddings	Asset pipeline
Perception	Sensor input, multimodal data	Cognitive substrate Layer 1
Meaning	Semantic extraction result	Cognitive substrate Layer 2
Thought	Deliberation record	Cognitive substrate Layers 3–10
Action	Tool call, memory write record	Execution engine
Evidence	Audit chain, compliance record	Kernel, SOE layer

Table 2.1: MemPacket content type variants and their producers

2.3 Content Addressing and CIDs

Every MemPacket is identified by its **Content Identifier (CID)**, computed deterministically from the packet’s content:

$$\text{CID} = \text{cls1-sha256-} \parallel \text{SHA256}(\text{canonical_json}(\text{content}))$$

The prefix **cls1-sha256-** identifies the Connector namespace and hash algorithm, making CIDs self-describing and forward-compatible.

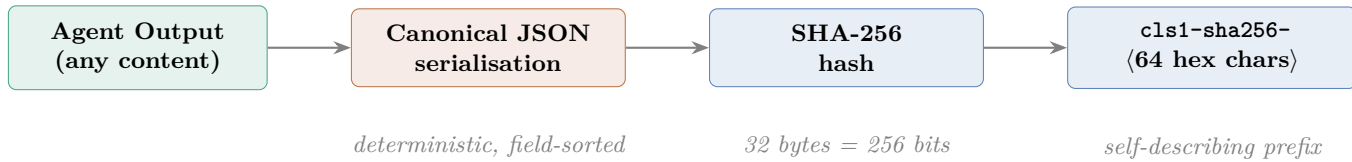


Figure 2.2: CID computation pipeline — content → canonical JSON → SHA-256 → CID

Four properties flow directly from this design:

1. **Deduplication** — Two packets with identical content share a CID. The storage layer never writes the same data twice.
2. **Integrity** — Any modification to content changes the CID. Corruption is detected without a separate checksum database.
3. **Independent verification** — Anyone with the content can recompute the CID and verify it matches, without trusting the server.
4. **Immutability** — MemPackets are write-once. Updating data means writing a new packet that references the old CID as its parent, creating an auditable lineage chain.

2.4 Namespace Hierarchy

The kernel organises memory into nine namespace types, each with distinct security levels and access semantics, analogous to Linux filesystem mounts:

Prefix	Type	Linux Analogy	Security	Purpose
/m/	Memory	/home	Standard	Private agent working memory
/k/	Knowledge	/usr	Protected	Shared, validated knowledge bases
/v/	Asset	/var	Tool I/O	Raw input files, staging area
/x/	App	/opt	Tool I/O	Executable apps via MCP
/c/	Core	/bin	Control	Native tools via CNP
/a/	Agent	/proc	Control	Agent control plane
/t/	Tool	/tmp	Tool I/O	Ephemeral tool I/O
/s/	System	/sys	Kernel	Kernel-reserved, read-only
/p/	Public	/srv	Public	Announcements, shared outputs

Table 2.2: Memory namespace types and their security boundaries

2.5 Syscall Interface

The kernel exposes over 60 syscalls organised into eight categories. Rather than raw code, what matters is the contract each category provides:

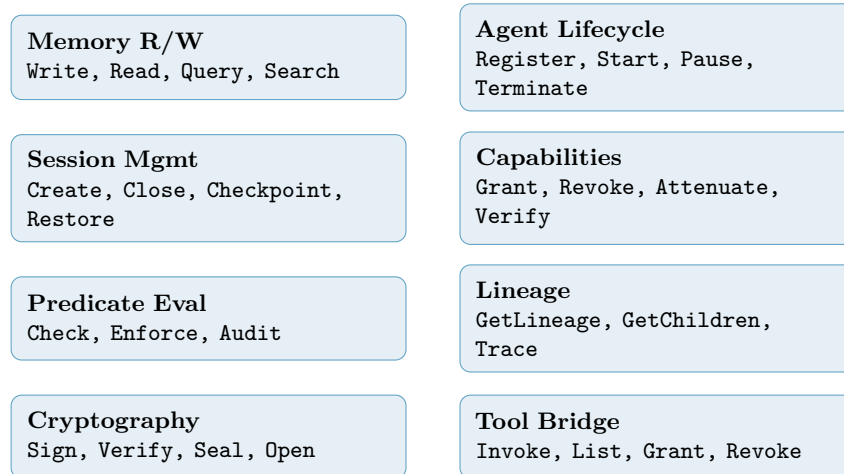


Figure 2.3: Eight syscall categories exposed by the Memory Kernel

2.6 Agent Lifecycle

Every agent in Connector passes through a well-defined state machine. The kernel enforces these transitions — no agent can execute outside an **Active** state, and no agent exits **Active** without a checkpoint.

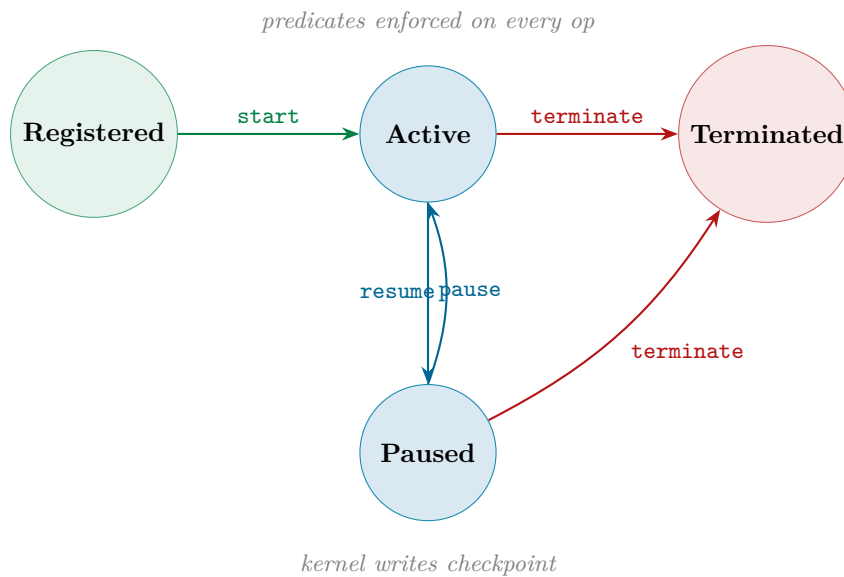


Figure 2.4: Agent lifecycle state machine enforced by the Memory Kernel

A key constraint: the transition from **Active** to any other state *always* triggers a checkpoint write. This means that even an unexpected crash (which skips the **terminate** transition) leaves the kernel able to restore the agent to its last checkpoint on the next boot.

2.7 Session Management

An **Agent** is a long-lived identity; a **Session** is a scoped execution context within that agent. Think of the agent as a Unix process and the session as a single system call batch — isolated, budgeted, and auditable.

Session Contract

A session is created with three inputs, and produces one output:

- **Budget** — Maximum tokens, memory bytes, tool calls, wall time
- **Predicates** — Governance rules that must hold throughout execution
- **Agent DID** — The identity executing within this session

Output: a signed **ExecutionReceipt** containing the full operation log, final state CID, and budget consumption.

The session lifecycle is: **Create** → **Execute** → **Checkpoint*** → **Close** → **Receipt**. Checkpointing can occur multiple times during execution. If the node crashes mid-session, the session can be restored from the last checkpoint.

2.8 The 11-Layer Predicate Chain

Before any kernel operation executes, the request passes through eleven enforcement layers in sequence. If any layer rejects the operation, execution stops immediately and an audit entry is written.

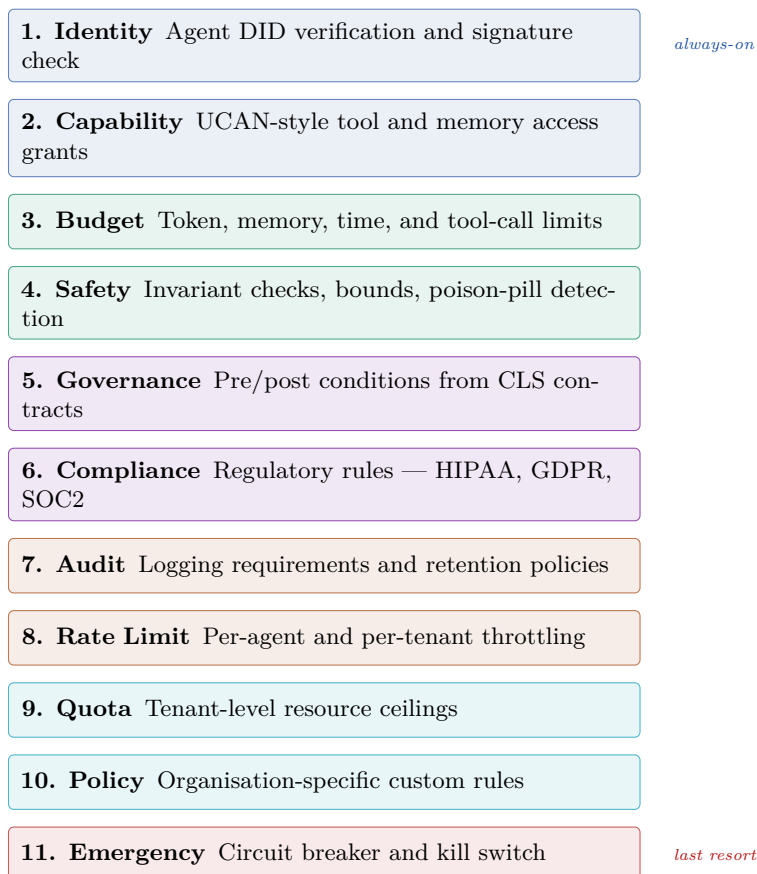


Figure 2.5: 11-layer predicate enforcement chain — evaluated in order for every kernel operation

2.9 Cryptographic Receipt Chain

Every session close produces a signed **ExecutionReceipt**. Receipts are themselves stored as **Evidence MemPackets**, so the full history of any agent's execution is a chain of content-

addressed, signed receipts — each referencing the CID of the previous one.

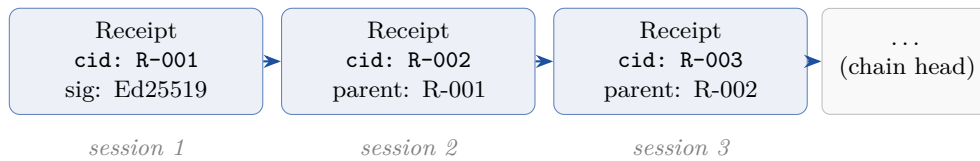


Figure 2.6: Execution receipt chain — each receipt CID-references its predecessor, forming a tamper-evident audit log

To verify that an agent’s history has not been tampered with, an auditor needs only the chain head CID and the raw content of each receipt. No trusted third party is required.

2.10 Storage Backends

The kernel abstracts over multiple storage backends through a single `MemoryStore` trait. All backends provide the same read/write/query semantics; only durability and scale characteristics differ:

Backend	Use case	Characteristics
SQLite	Local development, single-node	Zero-config, embedded, ACID
PostgreSQL	Production, multi-node	Full replication, WAL, high throughput
S3-compatible	Large binary content	Object storage, unlimited scale
IPFS	Decentralised deployments	Content-addressed natively, p2p

Table 2.3: Supported storage backends — all implement the same `MemoryStore` interface

Chapter 3

Execution Engines

3.1 Two Engines, One Platform

Connector provides two distinct execution engines, designed for complementary problem spaces. Understanding when to use each one — and how they interoperate — is central to building effective agents on the platform.

	Cognitive Substrate	CLS Runtime
What it is	An 11-layer deliberative reasoning pipeline inspired by cognitive architectures (BDI, ACT-R, Soar)	A contract compiler and VM for governed, structured agent logic
Best for	Open-ended reasoning, planning under uncertainty, multi-step deliberation	Deterministic workflows, compliance enforcement, auditable structured execution
Input	Raw perception (text, sensor, multi-modal)	A compiled SolutionContract (CID-addressed, signed)
Output	Commitment + action plan + reflection record	Signed ExecutionReceipt with full operation trace
Governance	Checkpointed at each of the 11 layers	Pre/post/invariant predicates enforced by the VM

Table 3.1: Cognitive Substrate vs. CLS Runtime — when to use each engine

Both engines write all intermediate and final state to the Memory Kernel as MemPackets, ensuring that every reasoning step and every contract execution is auditable and recoverable.

3.2 Cognitive Substrate

The Cognitive Substrate is Connector’s reasoning engine. It transforms raw inputs — text, sensor data, API responses — into committed, explainable actions through an eleven-layer pipeline. Each layer produces a typed output that becomes the input of the next, and each output is stored as a MemPacket with its own CID.

Why a pipeline, not a loop?

Most LLM agent frameworks use a simple observe → think → act loop. This works for simple tasks but breaks down when agents need to explain their reasoning, recover from mid-task failures, or satisfy compliance requirements. The 11-layer pipeline makes each reasoning *stage* explicit, checkpointed, and independently verifiable.

3.2.1 The 11-Layer Thought Pipeline

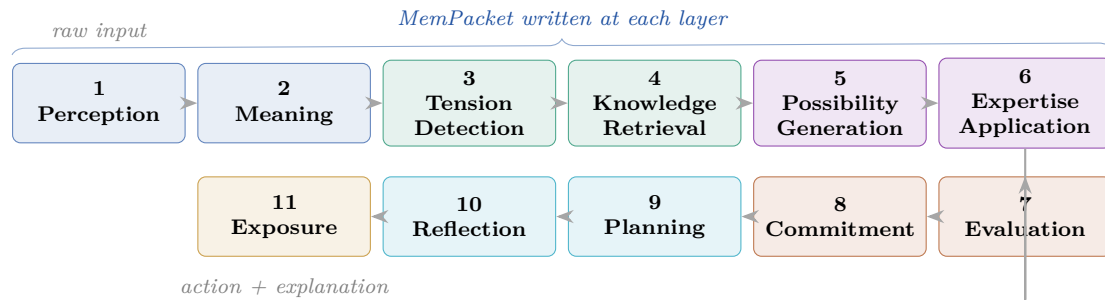


Figure 3.1: Cognitive Substrate 11-layer thought pipeline — each step is checkpointed as a MemPacket

#	Layer	What it does	Output type
1	Perception	Normalises raw input (text, sensor, multimodal) into a typed object	<code>PerceptionObject</code>
2	Meaning Extraction	Extracts entities, relations, intent from the perception	<code>MeaningObject</code>
3	Tension Detection	Identifies unresolved pressures — goals vs. reality gaps	<code>Tension[]</code>
4	Knowledge Retrieval	Queries memory for relevant facts (8 knowledge forms)	<code>KnowledgeSet</code>
5	Possibility Generation	Generates candidate actions from tensions + knowledge	<code>Possibility[]</code>
6	Expertise Application	Domain kernel scores each possibility (medical, finance...)	<code>EvaluationScores</code>
7	Evaluation	Multi-dimensional scoring: feasibility, impact, risk, cost	<code>RankedPossibilities</code>
8	Commitment	Selects best possibility; records revision history	<code>Commitment</code>
9	Planning	Builds a DAG of operations to execute the commitment	<code>ExecutionPlan</code>
10	Reflection	Compares expected vs. actual outcomes; extracts learnings	<code>ReflectionRecord</code>
11	Exposure	Renders result for human, peer agent, auditor, or debugger	<code>ExposurePacket</code>

Table 3.2: Cognitive Substrate layer descriptions

3.2.2 Expertise Kernels

Layer 6 is domain-aware. Rather than evaluating possibilities generically, the Cognitive Substrate delegates to a pluggable **ExpertiseKernel** that applies domain-specific rules. Built-in kernels include:

- **General** — Default scoring without domain constraints
- **Medical** — HIPAA-aware; flags PHI access, requires consent verification
- **Financial** — Compliance-aware; checks regulatory limits and disclosure rules
- **Legal** — Privilege-aware; enforces attorney-client boundary rules

Custom expertise kernels implement a single trait and are registered at boot time.

3.3 CLS: Connector Logic Standard

CLS is the *contract system* of Connector: a language for declaring agent logic, a compiler that transforms declarations into governed executables, and a runtime VM that executes them with full predicate enforcement.

The key insight of CLS is that agent logic should be declared, compiled, and verified *before* it runs — not interpreted at runtime from prompts or configuration files. A compiled contract is a content-addressed, signed artifact: it cannot be tampered with, and any execution of it is fully attributable.

3.3.1 CLS Compiler Pipeline

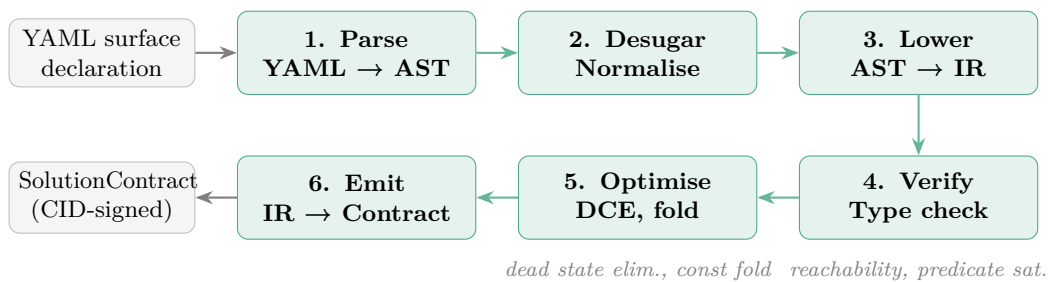


Figure 3.2: CLS 6-pass compiler pipeline — YAML declaration to signed SolutionContract

Pass 4 (Verify) is the most important: the compiler checks that all predicates are satisfiable, all state transitions are reachable, and all tool requirements can be resolved. A contract that fails verification is *never* stored and *never* executed.

3.3.2 SolutionContract Anatomy

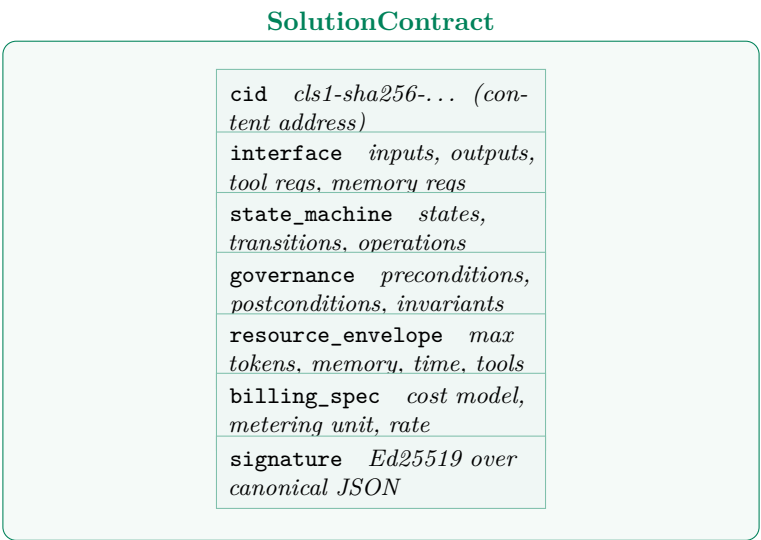


Figure 3.3: SolutionContract field anatomy — every field is included in the CID computation

3.3.3 Runtime Execution Lifecycle

When a contract is executed, the CLS runtime follows a deterministic seven-phase lifecycle:

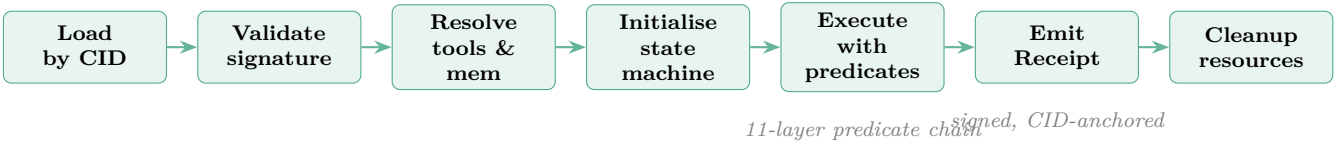


Figure 3.4: CLS runtime execution lifecycle — deterministic seven-phase flow

3.4 Resource Budgets

Both engines enforce a shared resource budget model. Budgets are specified when a session is created and enforced by the kernel in real time. Budget violations trigger immediate session termination, and the partial execution is still receipted — so operators can see exactly how much was consumed before the limit was hit.

Dimension	Unit	What it controls
max_tokens	tokens	Total LLM tokens consumed in the session
max_memory_bytes	bytes	Total MemPacket storage written
max_wall_time_ms	milliseconds	End-to-end session wall-clock time
max_tool_calls	count	Number of external tool invocations
max_memory_ops	count	Number of kernel read/write syscalls

Table 3.3: Resource budget dimensions enforced by both execution engines

Budget violation behaviour

When any budget dimension is exhausted, the session is terminated immediately. The kernel writes a final **BudgetExceeded** receipt with the exact consumption at time of violation. No partial results are silently discarded — the partial execution is fully traceable in the receipt chain.

3.5 Engine Integration Patterns

The two engines are designed to complement each other. Complex agent deployments typically use both:

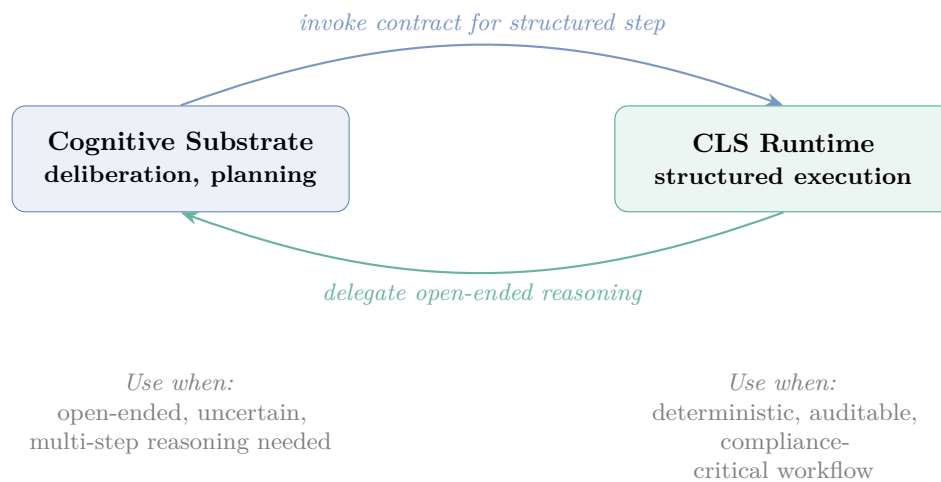


Figure 3.5: Engine integration — Cognitive and CLS engines delegate to each other as needed

Chapter 4

Protocol Layers

4.1 Why Three Protocols?

Agent communication in production spans three distinct contexts, each with different requirements:

- **Native intra-node** — Agents on the same Connector node need fast, capability-gated, typed messaging with full auditability. This is what **CNP** (Connector Native Protocol) is designed for.
- **Cross-platform** — An agent may need to communicate with an agent running on a different platform (LangChain, AutoGen, another Connector node). The **A2A** (Agent-to-Agent) bridge handles cross-platform interop.
- **External tools** — Agents need access to databases, APIs, search engines, and other services. The **MCP** (Model Context Protocol) bridge connects to any MCP-compatible tool server.

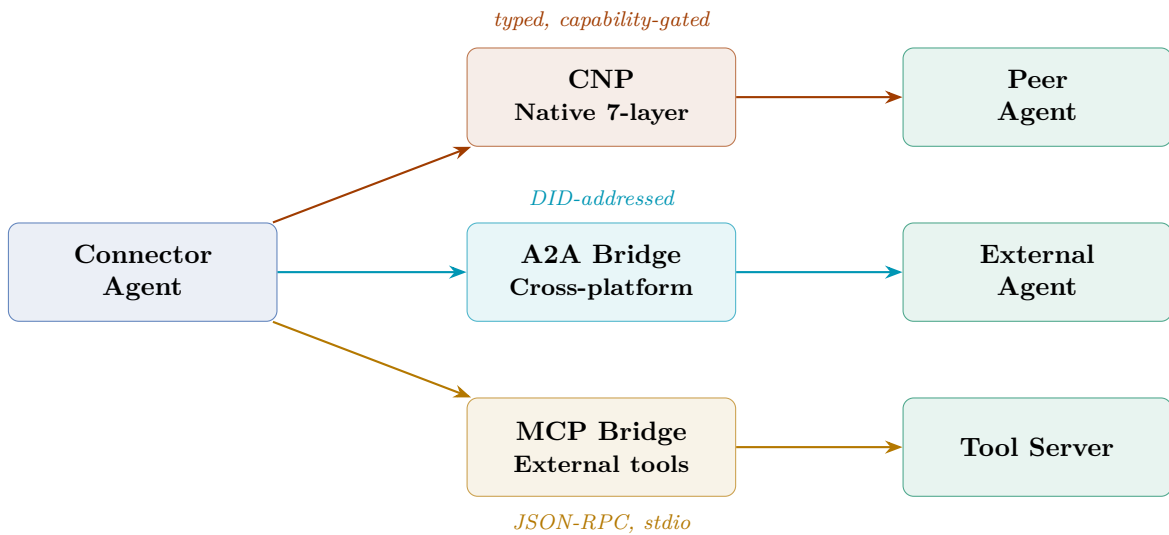


Figure 4.1: Connector protocol landscape — CNP (native), A2A (cross-platform), MCP (tools)

4.2 CNP: Connector Native Protocol

CNP is the primary communication protocol for agents running on the same Connector node. It is modelled on the OSI 7-layer model, but every layer is designed specifically for agent-to-agent communication: content addressing, capability gating, and cryptographic receipts are first-class concerns.

4.2.1 The 7-Layer Stack

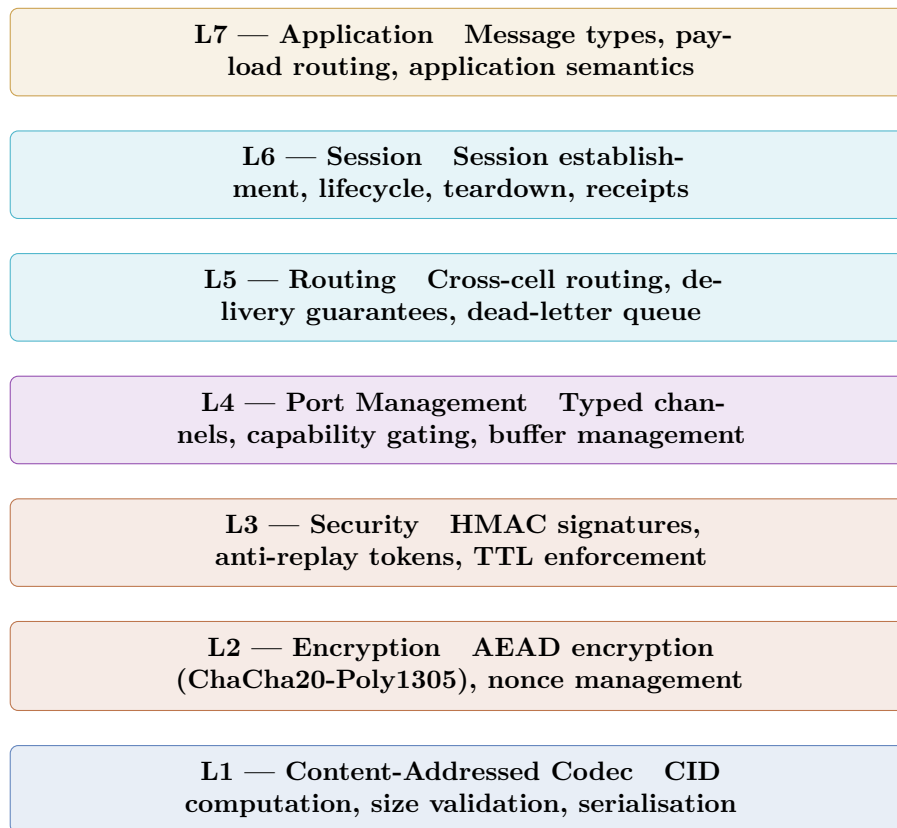


Figure 4.2: CNP 7-layer stack — modelled on OSI, purpose-built for agents

Every message passes through all seven layers on the way out, and all seven in reverse on the way in. Layer 1 gives tamper detection; Layer 2 gives confidentiality; Layer 3 gives authentication and replay prevention; Layer 4 gives capability enforcement; Layers 5–7 give routing, sessions, and application semantics.

4.2.2 Port-Based Communication

CNP does not use raw sockets or queues. All communication happens through **typed Ports** — named, capability-gated channels attached to an agent.

Port Type	Direction	Used for
Sensor	Inbound	Receiving raw sensor/environmental data
Actuator	Outbound	Sending commands to physical or virtual actuators
Memory	Both	Sharing MemPackets between agents
Tool	Outbound	Invoking tools and receiving results
Cognitive	Both	Passing cognitive messages between agents
Generic	Both	Untyped fallback for custom payloads

Table 4.1: CNP port types and their communication semantics

Opening a port requires a **Capability** token signed by the issuing agent. No capability, no port — the kernel rejects unauthorised connection attempts at Layer 4 before any payload is processed.

4.2.3 Capability Attenuation

Capabilities can be *delegated* (shared with another agent) but only with equal or reduced scope — never expanded. This UCAN-style attenuation forms a trust chain:

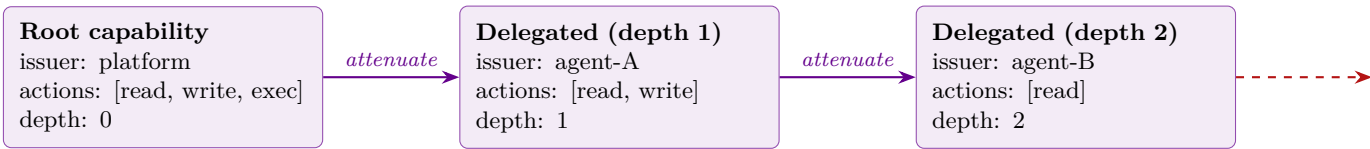


Figure 4.3: Capability attenuation chain — scope can only be reduced, never expanded; max depth is 3

4.2.4 Delivery Guarantees

CNP supports three delivery modes, selectable per message:

Mode	Retries	Behaviour	Use case
AtMostOnce	0	Fire and forget; no acknowledgement required	Telemetry, non-critical events
AtLeastOnce	∞	Retry until acked; possible duplicates	Commands requiring eventual delivery
ExactlyOnce	∞	Idempotent delivery with deduplication key	Financial transactions, state mutations

Table 4.2: CNP delivery guarantee modes

Undeliverable messages go to a per-agent **dead-letter queue**, where they can be inspected and replayed by an operator via `connectorctl`.

4.3 A2A: Agent-to-Agent Bridge

The A2A bridge allows Connector agents to communicate with agents on external platforms — LangChain, AutoGen, CrewAI, or other Connector nodes — without requiring CNP on both sides.

The bridge is **bidirectional**: Connector can receive inbound A2A messages and convert them to CNP for internal routing, and it can convert outbound CNP messages to A2A for delivery to external agents.

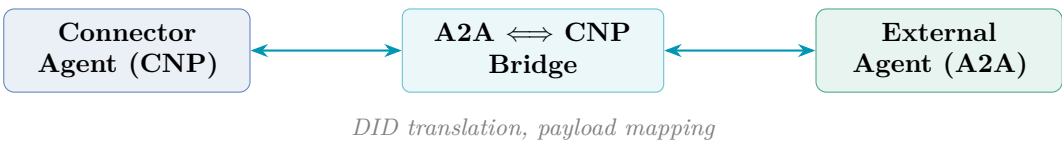


Figure 4.4: A2A bridge — bidirectional protocol translation between CNP and A2A

All inbound A2A messages are logged as MemPackets by the bridge, so even cross-platform communication is auditable from the Connector side.

4.4 MCP: Model Context Protocol Bridge

MCP is an open standard for connecting AI systems to external tool and data providers. Connector implements an MCP client bridge that allows any registered MCP server to be used as a tool by Connector agents.



Figure 4.5: MCP tool invocation flow — every call is wrapped in a MemPacket for audibility

MCP servers are configured at node startup. Each registered server exposes a list of named tools with JSON Schema input specifications. Agents address tools by name; the bridge handles the protocol translation transparently.

4.5 Protocol Selection Guide

Use case	CNP	A2A	MCP
Intra-node agent-to-agent messaging	✓		
Cross-platform agent communication		✓	
External tool / API integration			✓
Capability-gated typed channels	✓		
Typed sensor / actuator data	✓		
Cognitive reasoning message passing	✓	✓	
MemPacket sharing between agents	✓	✓	
Memory knowledge base queries	✓		✓
Cross-node cluster communication	✓	✓	

Table 4.3: Protocol selection guide — choose based on topology and capability requirements

4.6 Shared Security Properties

Regardless of which protocol is used, all three provide a common set of security guarantees enforced at the kernel level:

Property	How it is enforced
Authentication	Sender DID verified via Ed25519 signature on every message
Integrity	L1 CID detects any in-flight tampering; L3 HMAC prevents forgery
Confidentiality	L2 AEAD encryption (CNP); TLS for A2A and MCP
Non-repudiation	Signed delivery receipts stored as MemPackets
Audit trail	Every protocol operation logged to the Memory Kernel

Table 4.4: Security properties shared across all three protocol systems

Chapter 5

Developer Experience

5.1 The Glue Philosophy

Every powerful platform risks becoming inaccessible through its own complexity. Connector solves this with **Glue** — a unified developer surface that exposes the full power of the kernel, execution engines, and protocol stack through a simple, human-readable grammar.

Glue is not a wrapper or an SDK. It is a *language*: a set of verbs, nouns, selectors, and time flags that compose into commands with consistent, parseable output contracts. The same grammar works in the CLI, the REST API, and the Rust library — so operators, developers, and automated pipelines all speak the same language.

Three design principles of Glue

1. **Verbs first** — Commands read like instructions: `start agent`, `list agents`, `prove agent` — not `agent-start` or `agent.start()`.
2. **Hide complexity, not power** — One verb can orchestrate many kernel operations. The complexity is hidden, but not removed — it is accessible through flags.
3. **Instant value** — Every command produces useful output immediately, even with no flags. No ceremony, no configuration required to get started.

5.2 The Glue Grammar

Glue commands follow a consistent four-part structure:

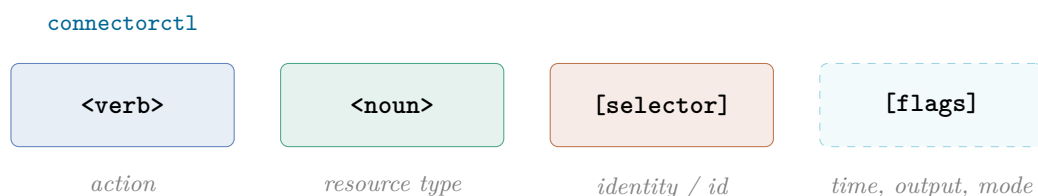


Figure 5.1: Glue command grammar structure

5.2.1 Verbs

Glue provides 12 primary verbs that cover the full agent lifecycle:

Verb	Category	What it does
<code>start</code>	Lifecycle	Start an agent, service, or node
<code>stop</code>	Lifecycle	Stop an agent or service gracefully
<code>restart</code>	Lifecycle	Stop then start, preserving checkpoints
<code>list</code>	Query	List agents, sessions, memory, contracts
<code>show</code>	Query	Show summary state for one resource
<code>inspect</code>	Query	Deep inspection — forensic mode available
<code>trace</code>	Diagnostics	Decision trace with time-window filter
<code>prove</code>	Evidence	Generate cryptographic proof for an agent
<code>explain</code>	Evidence	Decision-first explanation (<i>why</i> an agent acted)
<code>deploy</code>	Operations	Deploy an agent manifest or contract
<code>backup</code>	Maintenance	Checkpoint memory to an archive
<code>restore</code>	Maintenance	Restore from a backup or checkpoint

Table 5.1: Glue’s 12 primary verbs

5.2.2 Nouns

Ten primary nouns represent the resources Glue operates on: `agent`, `session`, `memory`, `contract`, `tool`, `port`, `node`, `cluster`, `key`, `user`.

5.2.3 Time Flags

Time flags enable point-in-time and range queries against the immutable MemPacket history — a direct expression of the kernel’s content-addressed store:

Flag	Format	Example
<code>-at</code>	ISO 8601 or Unix timestamp	<code>-at 2026-03-22T14:00:00Z</code>
<code>-since</code>	ISO 8601 or relative	<code>-since 1h</code>
<code>-last</code>	Duration shorthand	<code>-last 15m</code>
<code>-before</code>	Timestamp or event name	<code>-before incident-001</code>
<code>-snapshot</code>	Checkpoint CID	<code>-snapshot cls1-sha256-...</code>

Table 5.2: Glue time flags — enable time-travel queries against the kernel’s history

5.2.4 Output Flags

Output format is controlled by a single flag. Every command produces consistent, parseable output in all modes:

Flag	Output format
(default)	Compact human-readable table
-summary	One-line status line
-ops	Operations-focused: costs, budgets, throughput
-forensic	Full audit trail with CIDs and signatures
-narrative	Prose explanation for human readers
-json	Machine-readable JSON for automation

Table 5.3: Glue output format flags — same command, different audiences

5.3 connectorctl: The CLI

`connectorctl` is the primary operator interface for Connector nodes. It is a single compiled binary that implements the full Glue grammar.

```

1 # Check node health
2 connectorctl health
3
4 # List all running agents
5 connectorctl list agents
6
7 # Show a specific agent
8 connectorctl show agent medical-assistant-1
9
10 # Trace decisions in the last 15 minutes
11 connectorctl trace agent medical-assistant-1 --last 15m
12
13 # Generate a cryptographic proof
14 connectorctl prove agent medical-assistant-1 --json
15
16 # Explain the last decision made by an agent
17 connectorctl explain medical-assistant-1
18
19 # Deploy an agent from a manifest
20 connectorctl deploy agent ./manifests/triage-agent.toml
21
22 # Inspect with full forensic detail
23 connectorctl inspect agent medical-assistant-1 --forensic

```

Listing 5.1: `connectorctl` example session

5.3.1 Standardised Output Contracts

Every `connectorctl` command produces output conforming to a stable contract. Operators can pipe output to `jq`, monitoring systems, or alerting pipelines without special-casing each command:

```

1 # Human-readable table (default)
2 connectorctl list agents
3   ID                STATUS    UPTIME    COST TODAY
4   medical-assistant-1  running  2h 15m    $0.42
5   data-pipeline-2     running  5h 42m    $1.18
6   audit-agent-3       paused   ---       $0.07
7

```

```

8 # JSON output for automation
9 connectorctl list agents --json
10 {
11   "agents": [...],
12   "metadata": { "request_id": "req-abc", "duration_ms": 12 }
13 }

```

5.4 The REST API Surface

Every Glue verb is available over HTTP at `http://localhost:9091/api/v1/`. The URL structure mirrors the CLI grammar: `verb/noun` → endpoint path.

Method	Path	Action
GET	<code>/agents</code>	List all agents
POST	<code>/agents</code>	Register a new agent
GET	<code>/agents/:pid</code>	Show agent state
POST	<code>/agents/:pid/start</code>	Start agent
POST	<code>/agents/:pid/chat</code>	Invoke agent (LLM gateway)
POST	<code>/agents/:pid/decisions</code>	Record a governance decision
GET	<code>/agents/:pid/audit/receipts</code>	Audit receipt chain
GET	<code>/agents/:pid/cost</code>	Token cost summary
GET	<code>/agents/:pid/books</code>	Trust score and integrity report
GET	<code>/disputes/:id/report</code>	Compliance report for a decision
GET	<code>/disputes/:id/defense-package</code>	Full defense package
POST	<code>/proofs</code>	Generate cryptographic proof
GET	<code>/actionlog/regulation-report/:reg</code>	HIPAA/SOC2/GDPR report
GET	<code>/health/live</code>	Liveness probe
GET	<code>/health/ready</code>	Readiness probe

Table 5.4: Core REST API endpoints — the full Glue surface over HTTP

All responses follow a consistent envelope:

```

1 {
2   "ok": true,
3   "data": { ... },
4   "error": null,
5   "metadata": { "request_id": "req-xyz", "duration_ms": 8 }
6 }

```

5.5 Configuration Hierarchy

Glue configuration resolves through four layers, with later layers overriding earlier ones. This means sensible defaults work out of the box, and individual commands can always be overridden without changing global config:

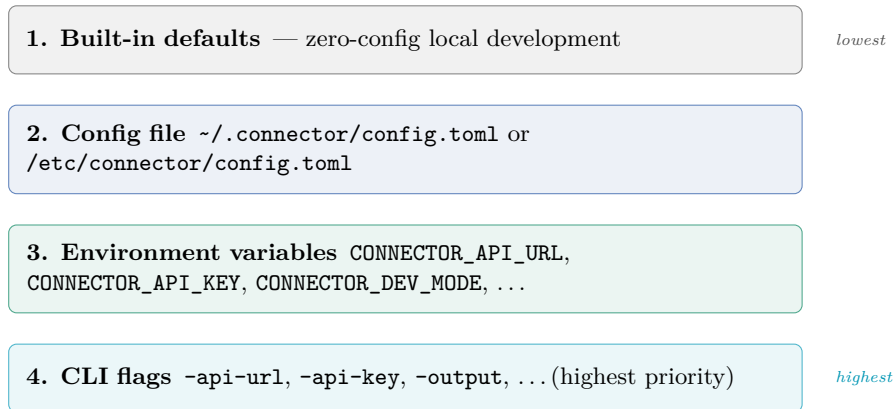


Figure 5.2: Configuration resolution order — later layers override earlier ones

5.6 Structured Error Model

Glue errors are always structured and actionable. Every error carries a machine-readable code, a human-readable message, and an operator hint:

```
1 {  
2   "ok": false,  
3   "error": {  
4     "code": "AGENT_NOT_FOUND",  
5     "message": "Agent 'medical-assistant-1' not found",  
6     "hint": "Run 'connectorctl list agents' to see available agents",  
7     "see_also": ["https://docs.connector.dev/agents/lifecycle"]  
8   }  
9 }
```

Error codes are namespaced (`AGENT_*`, `SESSION_*`, `BUDGET_*`, `AUTH_*`) so automated systems can pattern-match without parsing message text.

Chapter 6

Business Operations

6.1 Architecture: Two Servers

Connector separates execution from control. The **Platform Server** runs on the customer’s infrastructure and executes agents; the **License/Control Server** runs on Connector’s hosted infrastructure and manages accounts, licensing, and billing. The two servers communicate only for authentication and telemetry — no agent data or MemPackets ever leave the customer’s node.

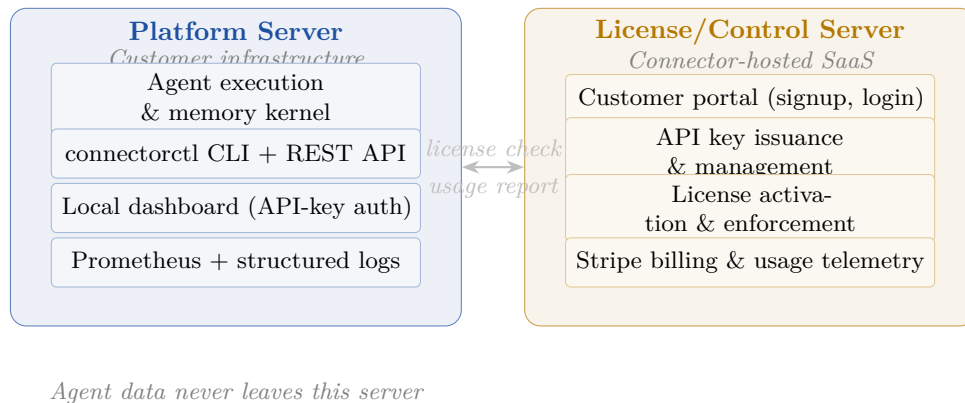


Figure 6.1: Two-server architecture — execution is local; control and billing are hosted

This separation is a deliberate **data sovereignty design**: compliance-sensitive organisations (healthcare, finance) can run agent workloads on-premise while still benefiting from the hosted portal, admin panel, and billing infrastructure.

6.2 Commercial Thesis

Connector should be sold as a **governed runtime and evidence layer for production AI**, not as a replacement for every orchestration, storage, observability, and compliance tool in the stack. The commercial message must stay narrow at entry and only broaden after trust is earned.

Avoid the platform-first trap

Do not lead with “Connector replaces your whole AI stack.” The better entry message is:

“When an AI workflow goes wrong, Connector shows what happened, why it happened, which policy allowed it, and which receipt proves it.”

6.2.1 Why now

The 2025–2026 market is favourable for a focused infrastructure vendor that helps enterprises move AI from experimentation into governed production. Menlo Ventures reports that enterprises now **buy more AI than they build**, with purchased solutions materially outpacing internal builds, and that AI deals reach production at higher rates than traditional SaaS. At the same time, investor guidance for 2026 remains traction-first: paid pilots, capital efficiency, and a credible next-round plan matter more than ambitious platform narratives.

6.3 Product Positioning

Core product definition — do not change this

Connector = AI Decision Audit, Proof, and Cost Control Layer

“When your AI makes a decision, Connector shows what happened, why it happened, proves it cryptographically, and controls the cost — in real time.”

6.3.1 Three Pillars

Three capabilities. Not a platform narrative. Not a suite. Three specific things a buyer can hold in their head and explain to their own team.

#	Pillar	What it means	Why a buyer pays for it
1	Audit	Every agent decision produces a trace: what happened, in what order, with what data, at what time. Tamper-evident. Kernel-level, not app-level.	“When something went wrong, I can replay exactly what the agent did and why. I do not need a developer in the room to explain it.”
2	Proof	Each trace is signed with a cryptographic receipt (CID-addressed, verifiable). The signature cannot be faked or retroactively edited. Structured for compliance output: HIPAA, SOC2 CC6/CC7, FCA.	“I can show this to my auditor, my regulator, or my legal team. It is not a log. It is signed evidence.”
3	Cost Control	Every agent action records token usage, compute spend, and API calls. Hard caps are enforced at the kernel before the action executes, not after. Budget per agent, per session, per team.	“I know exactly what each AI workflow costs. I can set a hard limit and it will not be exceeded. No surprise bills.”

Table 6.1: The three pillars of Connector — Audit, Proof, Cost Control

6.3.2 Entry Wedge

The correct wedge is not “replace all AI infrastructure.” It is to land on one specific painful workflow, deliver all three pillars against it, and then expand.

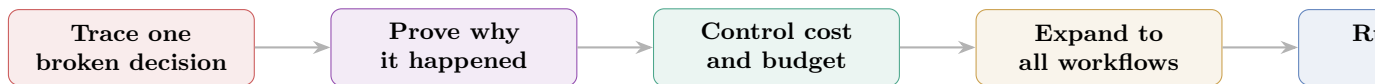


Figure 6.2: Commercial wedge — land on one broken decision, expand to runtime of record

6.4 Who Actually Pays: Reality-First ICP

First principle — lock this

Not “AI startups.” Not “customer support SaaS.”

Teams where AI is already making decisions that affect money or customers — and those decisions are going wrong or unclear right now.

If they are not feeling it weekly, they will not open their wallet.

6.4.1 The Three-Question Qualifier (Use Before Every Call)

All three must be true or move on:

1. **AI is live in production** with real users making real decisions — not a test, not a prototype, not “we are exploring.”
2. **Decisions affect money or customer trust** — refunds issued, transactions blocked, tickets actioned, approvals made automatically.
3. **They have already felt it go wrong** — a wrong refund they could not explain, a cost spike, an enterprise client who asked for an audit log they could not produce.

Disqualify fast — your time is the constraint in 2026

No prior AI incident = no urgent budget. Ask in the first 2 minutes: “*Has your AI ever made a decision you could not explain to a customer or your team?*” No = end the call. Yes = keep going.

6.4.2 Segment 1: Ecommerce AI Support — Enter Here First

Exact type: Shopify-focused SaaS, DTC automation tools, helpdesk products with AI automation. Companies whose product *is* AI-driven customer ops — refunds, returns, order edits, discount decisions.

Dimension	Concrete detail
Real pain	“We lost \$4K this week from wrong refunds.” “AI is too aggressive on discounts.” “A customer escalated and we have no idea why it issued that refund.” Happens weekly.
Who pays	Founder or Head of Ops (feels the financial pain). CX lead (faces escalations). CTO signs.
Pitch	<i>“Your AI is issuing refunds you cannot explain. We give you the full trace, the reason, and the proof of every decision — so you stop losing money and explain instantly.”</i>
Demo	AI issued a \$120 refund incorrectly. Show: trace (what happened), explain (which rule applied), prove (signed receipt), cost (token usage). 5 minutes. One bad decision. Buyer understands immediately.
Sales cycle	2–4 weeks. One buyer. No compliance review. High urgency.
Why \$10K–\$20K / 6 months is easy to approve	Processing 500 refunds/month at \$80 average = \$40K/month in refund volume. Even 3% wrong = \$1,200/month of bad money out. Debugging each incident manually = 3–4 engineer hours at \$150/hr × 15 incidents = \$6,750/month hidden cost. Total problem cost: \$8K–\$20K/month. Connector at \$2K–\$3K/month to fix it. The maths approves itself in 10 minutes.

Table 6.2: Segment 1 — Ecommerce AI Support: best wedge, fastest close, clearest ROI

6.4.3 Segment 2: AI Support Agent Startups — Action-Takers Not Chatbots

Exact type: Startups building agents that *take actions* — close tickets, call APIs, issue credits, resolve workflows end-to-end. Not chatbots that answer questions. Agents that do things with consequences.

Dimension	Concrete detail
Real pain	“Customer complained the AI did the wrong thing and we spent two days figuring out why.” “Our biggest enterprise prospect asked for an audit log and we have nothing to show.” “We keep getting blocked in procurement over AI compliance questions.”
Who pays	CTO or Founder. Head of Product when an enterprise deal is at risk.
Pitch	<i>“Your enterprise customers will ask ‘why did your AI do this?’ before they sign. We give you that answer instantly, with a signed receipt you can put in front of their compliance team.”</i>
Sales cycle	3–6 weeks. CTO approves directly. One blocked enterprise deal creates immediate urgency.
Why \$10K–\$20K / 6 months is easy to approve	One blocked or lost enterprise deal due to lack of auditability = \$30K–\$150K ARR delayed or gone. Connector at \$3K/month unblocks that deal in the next sales cycle. 1 deal recovered = 5–10× the pilot cost. Additionally: the audit trace becomes a <i>sales feature</i> they show in their demo. “Here is the signed proof of every decision our agent made.” That line closes deals.

Table 6.3: Segment 2 — AI Support Agent startups: audit as sales feature and deal-unlocker

6.4.4 Segment 3: Fintech — Selective, Not All

Exact type: Fraud detection tools, payment monitoring SaaS, risk scoring platforms. **Not banks. Not early-stage. Not “exploring AI.”** Only startups where AI is already making real financial decisions in production today.

Dimension	Concrete detail
Real pain	“AI blocked a legitimate transaction and the customer left.” “We need explainability for our next compliance review.” “Regulators are asking how our AI makes decisions and we have no structured answer.”
Who pays	CTO (primary). Head of Risk.
Pitch	<i>“If your AI blocks or approves money, you need to prove why — to regulators, to customers, and to your own risk team. We give you a signed, verifiable record of every decision.”</i>
Sales cycle	4–10 weeks. Slower than Segment 1 but CTO can approve under \$10K directly. Compliance officer adds 2–3 weeks.
Why \$10K–\$20K / 6 months is easy to approve	One wrong AI decision resulting in a regulatory inquiry = legal review at \$300–\$600/hr, 10–40 hours minimum = \$3K–\$24K per incident. Manual compliance analyst reviewing AI decisions = 0.5 FTE at \$80K/year = \$3,300/month. Connector at \$3K/month replaces or eliminates both.

Table 6.4: Segment 3 — Fintech: regulatory urgency creates real forcing function; enter after 2 support wins

6.4.5 Segment 4: Internal AI Ops — Only After Incidents

Exact type: Companies using AI internally for approvals, workflows, or ops automation. **Only approach if they have already had an incident** — an AI action that cost them time, money, or internal credibility. Without an incident, urgency is low and the deal will stall.

Dimension	Concrete detail
Real pain	“AI did something unexpected last month and it took us three days to untangle.” “Cost is completely unpredictable.” “We don’t fully trust it yet but we’re running it anyway.”
Close probability	Low without a prior incident. If they can name a specific incident that cost them, it becomes medium.
Strategy	Do not outbound to this segment. If they come inbound or if they mention a specific incident, qualify and proceed. Otherwise skip.

Table 6.5: Segment 4 — Internal AI Ops: only viable if they have had a named incident

6.4.6 Who Will Not Pay — Save Your Time

Do not contact these company types — they will never close in 2026

- **Chatbot SaaS.** They answer questions, they do not take actions. No financial consequence from a bad response = no urgency to audit it.
- **Early AI startups with no production usage.** They will say “this is interesting, come back in six months.” They are not lying. They mean it. Move on.
- **Enterprises without active AI decisions.** Large companies exploring AI take 6–18 months from pilot approval to contract. You do not have that runway in 2026.
- **“Exploring AI” companies.** If their website says “we are integrating AI” or “AI-powered coming soon” — they are 3–6 months from feeling the pain. You want companies that already feel it.

6.4.7 The Pilot Offer — Say This Exactly

The offer — one workflow, 30 days, sharp scope

“Give us one AI workflow where decisions matter — refunds, approvals, or actions. In 30 days we will give you:

- *Full traceability of every decision in that workflow*
- *Explanation of every action the agent took and why*
- *Proof — a signed receipt you can show your compliance team*
- *Cost visibility and hard budget control per session*

One workflow. \$2K–\$5K/month. Six months. You will know in 30 days whether it is worth continuing.”

The 30-day promise is critical. It converts a 6-month commitment into a 30-day decision. The buyer does not need to commit to the full value — they commit to seeing one workflow traced. If it works (it will), they renew. You get 6 months of revenue from a 30-day promise.

6.4.8 How to Find the Right Companies

Signal	What it means	Strength
Website says “AI auto-resolves” or “AI handles X”	AI is in production taking actions	Strong
Job posting for LLM Engineer or AI Engineer	They are actively building or scaling AI in production	Strong
Case study: “We reduced support cost by X% using AI”	AI is live, decisions are happening, mistakes exist somewhere	Best signal
Product demo shows AI taking autonomous actions	They have an action-taking agent, not just a chatbot	Strong
LinkedIn: employees with title “AI Ops” or “Agent Engineer”	Company is past exploration phase	Medium
Trustpilot or G2 reviews mention “AI did wrong thing”	Public evidence of live AI decisions going wrong	Very strong

Table 6.6: Prospecting signals — how to identify companies where AI is already causing problems

The 20-5-1 outreach formula for the first pilot

Find 20 companies matching Segment 1 signals. Shortlist 5 that show the “best signal” (reduced cost by X using AI, or a public review about wrong AI action). Pitch only this one question: “*Are you able to explain every AI refund decision today?*” If the answer is no or hesitation — schedule the demo. Target: 1 pilot closed from 20 companies

contacted. That is the realistic close rate. If you contact fewer than 20, you are not doing outreach — you are waiting.

The only metric that matters in outreach

“Don’t sell Connector. Sell stopping AI mistakes where money is leaking.”
You need 3 companies where AI is already hurting them. Not 100 customers. Not a big market. Three companies. Find them. Close them. That is 2026.

6.5 Customer Signup and Login

6.5.1 Signup Flow

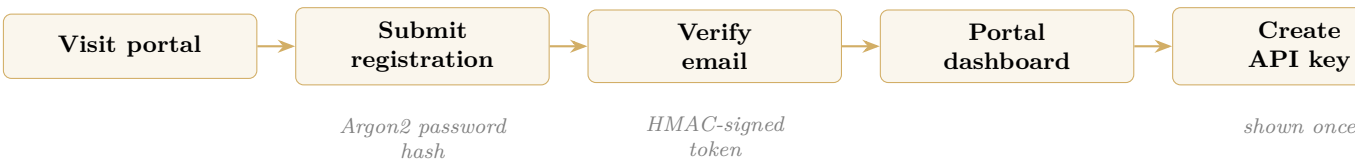


Figure 6.3: Customer signup and onboarding flow

6.5.2 Login Flow

Two separate login mechanisms serve two different contexts:

	Portal login	Dashboard login
Server	License/Control Server (hosted)	Platform Server (local node)
Credential	Email + password + optional TOTP	API key only
Token	JWT (24h expiry, refreshable)	API key (long-lived, revocable)
Access	Account management, billing, keys	Agents, sessions, memory, CLI

Table 6.7: Portal login vs. local dashboard login — different credentials, different contexts

6.6 API Key Management

API keys are the primary authentication mechanism for programmatic access to the Platform Server. They are issued by the License/Control Server and validated locally by the Platform Server — no round-trip to the hosted server is required for every request.

6.6.1 Key Lifecycle



Figure 6.4: API key lifecycle — plaintext is shown only once; only the hash is persisted

6.6.2 Key Structure

API Key Format

`ck_live_<32-char-random-bytes>`

- `ck_` — Connector key namespace
- `live_` or `test_` — Environment discriminator
- 32 random bytes (Base58 encoded) — 192 bits of entropy
- Prefix `ck_live_a1b2c3d4` visible for identification
- Full key stored as **SHA-256 hash only**

Keys carry a `scopes` list that limits what operations they can perform. Scope tokens follow the pattern `<noun>:<action>`, e.g. `agents:read`, `agents:write`, `decisions:write`.

6.7 License Activation

6.7.1 Activation Flow

After purchasing a plan, the customer runs a single command on their node:

```
1 connectorctl activate <license-key>
```

The node contacts the License/Control Server, which verifies the key, generates a machine fingerprint binding, and returns a signed license file. The license file is stored at `/etc/connector/license.key` and read at every boot.

6.7.2 License Tiers

The codebase currently contains more than one entitlement model. For business, sales, and documentation purposes, Connector should present one **canonical 2026 packaging model**: a small self-serve base, a team plan, an enterprise plan, and a paid pilot overlay used to land strategic accounts.

Dimension	Community	Pro	Team	Enterprise
Target buyer	Individual builder	Small production team	Cross-functional AI team	Regulated or large enterprise
Monthly price	Free	\$49–\$99	\$299–\$999	Custom / annual contract
Agents	2–3	Up to 20	High or uncapped	Contracted
Tokens	Small daily/monthly cap	500K monthly class	5M monthly class	Contracted or uncapped
Memory retention	30 days	1 year	Unlimited	Unlimited
HIPAA / advanced compliance	–	–	Optional add-on / controlled release	Included by contract
SSO / advanced admin	–	–	Optional	Included
On-prem license	–	–	Limited cases	Core offer
Support	Community	Email	Priority	Dedicated

Table 6.8: Recommended canonical packaging for Connector in 2026

6.7.3 Paid pilot overlay

Paid pilots should not be treated as ad hoc discounting. Internally, the control server already models pilots as **PilotGrant** overlays applied on top of a base entitlement. Commercially, this is the right design because it avoids creating one-off tiers for every prospect.

Pilot field	Commercial meaning
<code>tier_override</code>	Temporarily grant a higher plan for evaluation
<code>agent_limit_override</code>	Expand concurrency for one team or workflow
<code>packet_limit_override</code>	Expand memory / evidence retention for the pilot
<code>features_override</code>	Turn on gated features such as HIPAA, SSO, or multi-cell
<code>expires_at</code>	Force a commercial decision point instead of an endless trial

Table 6.9: Pilot overlays — commercial meaning of the internal pilot grant model

All license tiers and pilot overlays produce a signed **License** object with an Ed25519 signature. The Platform Server verifies this signature at boot and on every agent start without a network round-trip.

6.8 Billing Integration

Connector uses **Stripe** for all payment processing. The billing flow is fully event-driven through webhooks — the License/Control Server reacts to Stripe events to update entitlements in real time.

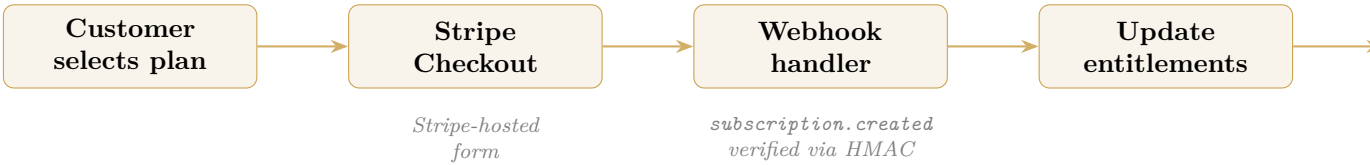


Figure 6.5: Stripe billing flow — webhook-driven entitlement updates

6.8.1 Handled Stripe Events

Stripe event	Connector action
<code>customer.subscription.created</code>	Upgrade tier, issue license key
<code>customer.subscription.updated</code>	Adjust agent/memory/token limits
<code>customer.subscription.deleted</code>	Downgrade to Free; agents above limit paused
<code>invoice.payment_failed</code>	Notify customer; 7-day grace period
<code>invoice.payment_succeeded</code>	Clear any payment-failure flag

Table 6.10: Stripe webhook events and their Connector side effects

6.9 Usage Tracking and Telemetry

The Platform Server reports a minimal telemetry payload to the License/Control Server every 15 minutes. This payload contains only aggregate counters — no agent data, no MemPacket content, no user queries.

Metric	Description	Used for
<code>agents_running</code>	Active agent count at time of report	Limit enforcement
<code>memory_packets</code>	Total MemPacket count	Storage billing
<code>tokens_consumed</code>	Cumulative LLM tokens this period	Token budget billing
<code>uptime_seconds</code>	Node uptime	SLA calculation
<code>session_count</code>	Sessions opened this period	Usage graphs
<code>tool_invocations</code>	External tool calls this period	Usage graphs

Table 6.11: Telemetry metrics reported to the License/Control Server — aggregates only

These metrics power the customer dashboard usage graphs, admin-panel fleet analytics, and (in future releases) usage-based pricing for token consumption above tier limits.

6.10 Go-to-Market Motion

Connector should use a **hybrid bottom-up + top-down motion**. The initial user is usually technical: a platform engineer, AI engineer, or Head of AI who needs to explain or control one production workflow. The deal closes when that technical win becomes legible to a broader stakeholder set: compliance, security, and the executive sponsor responsible for rollout risk.

Motion layer	Primary goal	Recommended execution
Bottom-up	Create urgency and technical conviction	CLI demo, local node install, one broken workflow explained end-to-end
Founder-led sales	Convert interest into a paid pilot	Direct outreach to 30–50 target accounts in healthcare, fintech, and enterprise AI platform teams
Top-down validation	Remove deal blockers	Bring in CTO, compliance, and security reviewers early with architecture and data-sovereignty materials
Land-and-expand	Grow from one workflow to one business unit	Add quotas, policy control, audit evidence, and more workloads after pilot success

Table 6.12: Recommended GTM motion for Connector

6.10.1 Why this motion fits the product

Menlo Ventures’ 2025 enterprise AI findings matter here: AI products are being purchased more often than built, they move to production at unusually high rates once evaluation starts, and a meaningful share of AI spend is initiated bottom-up. Connector should therefore **land technically and close organizationally**.

6.11 2026 Entry Plan

6.11.1 Primary commercial objective

The 2026 objective should be to prove that Connector is not merely a compelling architecture, but a repeatable commercial wedge. A practical target is:

2026 core target

10 paid pilots, each priced at roughly \$5k–\$7k MRR equivalent, with at least 4 pilot-to-annual conversions by year end.

In practice, that pilot price can be packaged as a **90-day paid pilot at \$15k–\$21k total**, which preserves the MRR-equivalent narrative while fitting enterprise procurement more naturally than a pure month-to-month experimental contract.

6.11.2 Recommended target account profile

The first 10 pilots should not be spread across every possible industry. They should be concentrated where Connector's wedge is strongest:

- **4 healthcare / healthtech pilots** for patient operations, triage, prior authorization, or compliance-heavy workflows.
- **3 fintech / insurtech pilots** where decisions must be explainable to internal auditors or customer support teams.
- **3 enterprise platform pilots** where internal AI platform teams are trying to standardize agent operations across multiple teams.

This focus improves message discipline, sales efficiency, and referenceability.

6.11.3 Pilot structure

Pilot element	Recommended structure
Duration	8–12 weeks, with explicit success criteria signed before kickoff
Commercial form	\$15k–\$21k fixed pilot fee (equivalent to \$5k–\$7k MRR)
Scope	One workflow, one team, one production-like environment
Success criteria	Show one failure chain, one policy decision path, one signed evidence trail, one operator workflow improvement
Technical deliverable	Connector node deployment, portal account, API keys, pilot overlay entitlement, usage dashboard
Conversion path	Annual Team or Enterprise contract with expansion to more workflows or business units

Table 6.13: Recommended paid pilot design

6.11.4 Quarter-by-quarter execution plan

Period	Execution objective
Q1 2026	Finalize narrative, tighten demo, ship stable onboarding, identify 30–50 target accounts, close first 2 paid pilots
Q2 2026	Reach 4–5 paid pilots, publish 2 detailed case studies or design-partner stories, refine packaging and pricing based on objections
Q3 2026	Reach 7–8 paid pilots, convert first 2–3 pilots to annual contracts, formalize repeatable founder-led sales process
Q4 2026	Reach 10 paid pilots total, convert at least 4 to annual contracts, use the data room to support pre-seed or seed fundraising

Table 6.14: 2026 execution plan — from wedge validation to repeatable GTM

6.12 Fundraising Narrative

6.12.1 What the pre-seed story should be

The pre-seed story is not “we built a very deep AI operating system.” The pre-seed story is:

Fundraising message discipline

Connector helps enterprises move AI into production by giving them a governed runtime and evidence layer. The company has a clear entry wedge, paid pilot traction, strong expansion logic, and a product architecture aligned with data sovereignty requirements.

Founder Institute’s 2025 benchmark guidance is useful here: in 2026 investors want paid traction, capital efficiency, and a concrete plan for what the next round of capital unlocks. For Connector, that means the fundraise should be anchored in **evidence of demand plus clarity of use of proceeds**.

6.12.2 Recommended pre-seed milestones

Milestone area	What investors should see
Commercial traction	4–6 active paid pilots minimum; 10 is a strong year-end target
Revenue quality	Clear pilot pricing discipline and visible path to annual contracts
Category clarity	One-sentence wedge buyers immediately understand
Referenceability	At least 2 strong customer stories in regulated or high-consequence workflows
Capital plan	Hiring and roadmap tied to specific go-to-market and product bottlenecks

Table 6.15: Milestones that support a credible pre-seed raise

6.12.3 Use of proceeds

If Connector raises a pre-seed round, the capital should primarily fund:

- **Product hardening** — onboarding, pilot tooling, entitlement consistency, dashboards, packaging polish.
- **Customer success and implementation** — fast pilot launches and clean conversions.
- **Founder-led GTM leverage** — tighter outbound process, repeatable demo assets, sector-specific collateral.
- **Compliance readiness** — the materials and controls required for enterprise trust in healthcare and fintech sales cycles.

6.13 Commercial Scorecard

The business should be managed against a small number of metrics rather than a large vanity dashboard.

Metric	Why it matters	2026 target
Paid pilots closed	Proves the wedge is commercially real	10
Pilot MRR-equivalent	Shows pricing discipline and revenue quality	\$50k–\$70k combined
Pilot-to-annual conversion	Shows the product expands beyond curiosity	40–50%
Average pilot launch time	Shows operational readiness	Under 30 days from signature
Champion-to-exec progression	Shows multi-stakeholder sale health	In every paid pilot
Gross logo churn	Protects early narrative quality	Near zero during pilot year

Table 6.16: Commercial scorecard for the first year of GTM

6.14 SWOT Analysis

Dimension	Connector assessment
Strengths	Strong technical differentiation, data-sovereign architecture, runtime-native governance, explainability with proof, clear value in regulated workflows
Weaknesses	Broad architecture can be mis-pitched, category education is still required, enterprise trust must be earned, packaging needs tight canonicalization
Opportunities	Enterprise AI budgets are growing, buyers increasingly purchase instead of build, agent governance is still early, regulated verticals need practical auditability
Threats	Larger incumbents may add partial governance features, category confusion may slow deals, long enterprise cycles can starve a small team, broad platform positioning can dilute the wedge

Table 6.17: SWOT summary for Connector’s 2026 business posture

One sentence to keep the company honest

Connector should not try to replace the world before it earns trust in one workflow. The product wins when it proves one broken AI decision path, controls it, and then expands from that foothold.

Ground Truth: Pre-Revenue Founder Economics

The preceding sections of this chapter describe where the business should be. This chapter describes how two founders with no revenue actually get there. Every number here is drawn from real startup data, not investor storytelling.

The salary problem is a business problem

A founder who cannot pay rent cannot execute. This is not a personal weakness — it is a planning failure. If you go full-time without solving the salary question first, you will make desperate decisions (wrong customers, wrong pricing, wrong hires) within four months. Solve it in writing before day one of full-time work.

Founder Runway: How to Pay Two Salaries Before Any Revenue

The minimum monthly requirement for two founders to survive full-time is honest and non-negotiable:

Item	Details	Monthly cost
Founder 1 living wage	Rent, food, transport, health cover (modest)	\$3,500–5,000
Founder 2 living wage	Same basis	\$3,500–5,000
Tooling & infrastructure	GitHub, cloud credits, domain, legal templates	\$400–600
Travel for pilot meetings	1–2 in-person visits per quarter	\$500–800
Total monthly burn	Pre-revenue, two founders	\$8,000–11,000

Table 6.18: Realistic monthly burn for two founders before any revenue

Four mechanisms fund this. Choose one or combine them:

1. **Pre-seed SAFE from angels (\$75K–\$150K).** One or two angel investors who understand developer-infrastructure businesses. At a \$2M–\$3M cap SAFE, \$100K gives 12–15 months of founder salaries at minimal living rates. This is the cleanest option — no board seat, no preferred shares, fast to close. Angels at this stage are usually former founders, CTOs, or senior engineers at AI companies. They invest in people and in conviction about the problem, not in a product that works.
2. **One founder keeps a part-time consulting income.** One day per week of senior engineering consulting (\$1,200–2,000/day) = \$5,000–8,000/month from one founder. This extends runway indefinitely but costs roughly 20% of that founder's product

time. Use this as a bridge, not a permanent state. The ceiling is 3–4 months before it noticeably slows the company.

3. **First pilot revenue within 90 days.** Two pilot customers at \$2,000/month each = \$4,000 MRR. Combined with minimal savings, this covers both founder salaries. This is achievable but requires the founders to have built a working demo and a short list of warm contacts before going full-time. Do not go full-time and *then* start building the demo.
4. **Government and accelerator non-dilutive grants.** In the UK (Innovate UK), EU (EIC Accelerator), Canada (SR&ED), and USA (SBIR Phase I for AI infrastructure), grants of \$25K–\$150K are available for AI systems research with a novel component. Connector’s predicate-chain governance and CLS compiler are genuinely novel. Timeline: 3–6 months to apply and receive. Not fast enough to be the primary bridge, but valuable as a secondary source.

The honest decision tree

If you have at least 4 months of personal savings each *and* a working demo *and* a list of 20 warm contacts: go full-time and close pilot 1 within 90 days. If you do not have all three, line up the angel SAFE first. Going full-time without runway is not hustle — it is bad planning.

Who Actually Pays During Pilots

The commercial scorecard in the prior section targets \$50K–\$70K combined pilot MRR. That number is real but the path to it does not go through large enterprises.

Enterprise sales will not save 2026

Fortune 500 companies, large banks, and hospital systems have procurement cycles of 6–18 months. They require SOC2 Type II (9–12 months and \$30K–\$80K to obtain), a security questionnaire response that takes 2–3 weeks per account, an infosec review, a legal MSA, a DPA, and an SLA with financial penalties. Chasing these accounts in 2026 means zero revenue in 2026. Do not do it.

The ideal pilot customer in 2026 looks like this:

Dimension	Ideal profile
Company stage	Funded AI startup (post-seed to Series A), or boutique AI consulting firm, or Series A/B SaaS company actively adding AI agent features
Company size	8–60 employees. Engineering team of 5–25.
Funding raised	\$2M–\$20M. Enough budget to move without a committee, not so large they need procurement.
Buying decision	CTO or VP Engineering makes the call alone or with one other person. No legal committee, no CISO review in round 1.
Technical profile	Team already using LLMs in production. Have already felt the pain of agents making decisions they cannot explain or audit.
Compliance pressure	At least one of: HIPAA, SOC2 (in progress), GDPR, or a regulated-industry client they are selling to.
Concrete examples	Early digital health startup building AI clinical decision support. AI-native fintech startup with LLM-driven underwriting. Boutique AI consulting firm that needs governance as a differentiator for their own client proposals. Series A developer-tools company adding AI features and worried about audit trails.

Table 6.19: Ideal pilot customer profile for 2026 — not enterprise, not consumer

Why These Customers Pay

The question is not just who pays — it is why they pay when Connector is early and the codebase is not yet fully polished.

1. **Build vs. buy arithmetic.** A funded startup burning \$200K/month on salaries would need 2–3 senior engineers for 12–18 months to build what Connector delivers. That is \$300K–\$600K in engineering cost. A \$2,000–\$3,000/month pilot is not a purchase decision — it is a cost-of-alternatives decision. The CTO’s mental model is: “could we build this ourselves?” When the answer is “yes but it would take us 18 months,” Connector wins.
2. **Design partner access.** At the \$1,500–\$3,000/month pilot price, the customer is not buying finished software. They are buying direct access to the founders who will build exactly the feature they need next. Founders respond to their Slack messages on the same day. Enterprise vendors do not. Early customers pay for that relationship — it is a real competitive advantage for them internally.
3. **Competitive differentiation.** In regulated or compliance-adjacent markets, a startup that can say “our AI agents produce a cryptographic proof of every decision” to their own enterprise clients wins deals that their competitors lose. The pilot customer is not buying Connector for themselves — they are buying it to use as a sales argument to *their* enterprise customers.
4. **Urgency.** Funded startups have a board. The board asks about AI governance now. The CTO has 6–8 weeks to have an answer. Connector is the answer they can deploy

in days, not months. That urgency converts to a paid decision without a procurement process.

10 Pilots, 2 People: The Actual Playbook

Ten paid pilots from two founders in 12 months is achievable. It requires a specific method, not general hustle.

The codebase condition is honest

As of early 2026, the Connector platform has excellent architectural depth (Memory Kernel, CLS compiler, CNP protocol, SOE) but lacks the operational maturity that makes self-service onboarding possible: a one-command deployment, complete API documentation, a stable demo environment, and polished error messages. This means the first 5 pilots must be **concierge-style** — the founders do the setup, hold the customer’s hand, and manually resolve friction. This is not a weakness to hide. It is the correct strategy: you learn 10× faster and customers feel 10× more valued.

Phase	Period	Target	Method	MRR
Phase 0	Jan–Feb 2026	Demo-ready	Fix deployment: Docker Compose single-node, 5-page integration doc, 3 worked examples, stable local demo that runs in under 10 minutes	\$0
Phase 1	Feb–Apr 2026	2 pilots	Direct outreach to 60–80 funded AI startups via LinkedIn, YC company list, Twitter/X. Founders do every demo personally. Concierge onboarding with weekly calls. Price: \$1,500–\$2,000/month	\$3K–\$4K
Phase 2	May–Jul 2026	5–6 pilots	Use Phase 1 learnings to write a one-page onboarding guide. Outreach to 100–150 targets with a social proof statement (“2 companies using this in production”). SDK hire joins to absorb integration support	\$10K–\$15K
Phase 3	Aug–Oct 2026	10 pilots	Semi-self-service. Referrals from Phase 1 customers. Sales hire joins to run outbound pipeline. Price increases to \$2,500–\$3,500/month as the product matures	\$22K–\$30K

Table 6.20: Phased pilot capture plan — 2 founders, concierge model, no enterprise

The outreach mechanics are not complex: identify companies by searching for “LLM agents”, “AI workflow”, or “AI governance” on LinkedIn job postings, AngelList, and Crunchbase. Companies actively hiring for these roles have the problem Connector solves. Message the

CTO directly. The message is two sentences: what you solve and who you have already helped. No deck. No brochure. A demo link that runs in the browser in 5 minutes.

The Three Hires: Who, Why, and When

Wrong hires destroy early startups faster than no hires

An enterprise AE costs \$120K–\$180K/year OTE and expects a polished product, a clean CRM, and a well-defined ICP. They will leave after 90 days if they cannot close. An RTC engineer or distributed-systems researcher costs \$160K–\$220K/year and will spend 6 months understanding the codebase before shipping anything customer-visible. Neither is the right hire in 2026. The right hires are people who can do multiple jobs badly rather than one job perfectly.

Hire	Role and profile	Timing	Monthly comp	Equity
Tech 1	Product engineer (Rust / full-stack). Reads the existing code-base, ships customer-visible features, handles deployments at pilot sites, helps debug integration issues. Has worked at a startup before. 5–8 years. Comfortable with	After pilot 3, MRR $\geq$ \$6K	\$5,500–7,500/mo	0.75–1.25%

Why Not Hire Earlier

Hiring before revenue reaches the thresholds above means the company is paying salaries that the pilots do not yet cover. That shortens the runway without accelerating the outcome. The two founders can operate:

- Pilot 1–2: Founder A owns product and architecture. Founder B owns outreach, demos, and onboarding. Both do customer calls.
- Pilot 3–5: One founder is now spending 60% of time on customer integration issues. That is the signal to hire Tech 1.
- Pilot 5–7: Integration issues are resolved by Tech 1 but SDK surface and onboarding repeatability is still fragile. That is the signal to hire Tech 2.
- Pilot 7–9: Outbound is now the bottleneck. The founder cannot run sales, manage three hires, and keep the product moving. That is the signal to hire Sales 1.

Honest Monthly Cash-Flow Projection

Month	Pilots	MRR	Monthly burn	Net monthly
1–3	0	\$0	\$9K (founders only)	–\$9K (funded by pre-seed or savings)
4–5	2	\$4K	\$9K	–\$5K (still drawing down)
6–7	4	\$9K	\$9K	±\$0 (cash-flow neutral)
8–9	6 + Tech 1 joins	\$15K	\$15K	±\$0
10–11	8 + Tech 2 joins	\$21K	\$20K	+\$1K
12	10 + Sales 1 joins	\$27K–\$30K	\$24K	+\$3K–\$6K cash-flow positive

Table 6.22: Honest 12-month cash-flow model — two founders, no external capital after pre-seed

At month 12 with 10 pilots at an average of \$2,700/month, Connector is at \$324K ARR with three hires and two founders generating a small positive monthly cash flow. This is not an ambitious projection — it is the **minimum** viable outcome from the 2026 plan. It funds the story for the Seed 1 round (\$1M–\$2M) at a valuation that reflects real revenue traction, not pre-revenue promise.

The goal of 2026 is not \$1M ARR

The goal of 2026 is to prove that a specific customer pays a specific price for a specific outcome — reliably, repeatedly, without the founders being in every call. That proof is worth more to a Seed investor than any amount of unpaid pilots or letters of intent. Eleven paying customers at \$2,500/month beats fifty LOIs every time.

What “Pilot-Ready” Means for This Codebase

The Connector codebase has architectural depth that most early-stage infrastructure companies take two additional years to reach. The Memory Kernel, CLS compiler, CNP protocol, SOE, and cognitive substrate are genuine technical assets. However, the gap between “architecturally correct” and “a customer can use this next Tuesday” is significant and must be closed before outreach begins.

Component	Current state	Gap to pilot-ready	Effort
Memory Kernel	CoreStable public API surface, 5 documented operations implemented, CID working		2–3 weeks
Deployment	Requires manual setup, multiple steps	Docker Compose single-node that works in one command	1–2 weeks
Demo environment	Interactive browser-accessible demo with a scripted walkthrough requires context to run		2 weeks
Python SDK	Not yet customer-facing	10-endpoint coverage, pip installable, 3 worked examples	3–4 weeks
API documentation	Internal only	151-page integration guide covering the 5 pilot workflows	2 weeks
Error messages	Technical / internal	Human-readable, actionable, no internal stack traces to customers	1 week
CLS compiler	Design complete, in development	A working compiler for 3 contract patterns used in pilot workflows	4–6 weeks
Billing & licensing	Architecture built	End-to-end test: sign up, activate, use, invoice	1–2 weeks
Total			8–12 weeks

Table 6.23: Gap analysis: current codebase state to pilot-ready

This 8–12 week window (January–March 2026) is **Phase 0**. No outreach happens during this phase. No demos. No sales conversations. The two founders build the demo environment, write the first integration guide, fix deployment, and get the Python SDK to a state where a CTO can try it in an afternoon without calling anyone. Only then does Phase 1 outreach begin.

Do not start selling before the demo works

Reaching out to 80 potential pilot customers before the demo runs reliably wastes the only asset you cannot recover: warm introductions. A CTO who tries the demo, hits an error, and does not get a response within 2 hours will never be a customer. You get one first impression per person. Use Phase 0 to make it count.

Fundraising Reality: The Honest Timeline

The fundraising timeline in the growth chapter requires correction against execution reality. Raising too early produces worse terms, wasted founder time, and investor relationships built on a story that has no proof yet.

Do not raise a pre-seed in 2026

Going to angels with zero paying customers produces a \$1–1.5M cap SAFE at best. Going to the same angels in Q4 2026 with \$15K–25K MRR and 8 paying customers produces a \$2.5–3.5M cap SAFE. That difference compounds forever: it means *less dilution for the same cash*. The right strategy is to bootstrap through 2026, accumulate traction, and raise on terms that reflect proof rather than promise.

Round	Timing	Pre-conditions required	Target raise	SAFE cap
Pre-seed	Q4 2026–Q1 2027	8–10 paying pilots. \$15K–25K MRR. Founders paid from pilot revenue. Working demo. Clear ICP proven in two verticals.	\$75K–125K	\$2.5M–3.5M
Seed 1	Q4 2027	15–20 paying customers. \$25K–40K MRR (\$300K–480K ARR). NRR above 100%. One reference customer willing to speak to investors. Team of 2 founders plus 2–3 hires. ICP is narrow and proven, not broad.	\$500K–1.5M	\$4M–6M post-money
Series A	Q3–Q4 2028	25–35 paying customers. \$600K–1M ARR. NRR	\$2M–5M	\$12M–20M post-money

Pre-Seed Reality: Why \$15–25K Angel Checks Are Hard

The pre-seed round consists of 3–5 individual angel checks of \$15K–25K each (total: \$75K–125K). On paper this sounds simple. In practice it is friction-heavy:

- Each angel requires a 30–60 minute call, a 5-slide company summary, a SAFE term sheet, and follow-up. With 10–15 conversations to close 4 cheques, the process takes 8–12 weeks of active founder time. During that period, one founder must keep running sales and product.
- Angels at this size often ask for MFN (Most Favoured Nation) provisions, pro-rata rights, or board observer seats. Agree only to pro-rata. The others add administrative overhead that kills momentum.
- The right angels are former founders or senior engineers at AI infrastructure companies (Databricks, Weights & Biases, Scale AI, Anthropic alumni). They invest in the *problem* conviction, not the product state. Reaching them requires warm introductions, not cold emails.
- Rule: if you spend more than 90 days trying to close the pre-seed, stop and return to pilot execution. Revenue is a better bridge than a stalled fundraiser.

Seed 1: What Investors Actually Need to See

A Seed investor in 2027 is not funding a demo. They are funding a pattern: a repeatable process that converts a specific type of customer into a paying contract. The proof they require is not *that* customers pay — it is *why* they renew.

At the Seed 1 conversation, the founding team must be able to say:

- “We have 15 customers. 12 have renewed past the initial 6-month pilot. 3 have expanded their contract.”
- “Our average pilot-to-annual conversion is 60%. Our NRR is 112%.”
- “Here is the exact workflow where we win: a Series A AI startup or mid-market SaaS company that is deploying LLM agents in a compliance-sensitive context and needs cryptographic audit trails.”
- “Here is the CTO who will take your call and explain exactly why they paid.”

Without these four statements grounded in real data, no credible Seed fund writes a cheque. This is why the 2026 pilot execution is not a “revenue target” — it is a **fundraising instrument**.

From Series A to Self-Sufficient by End of 2029

The Series A funds the scaling of a proven motion. It is not for experiments. The companies that fail between Seed and Series A spend the Seed money rebuilding the product for a new ICP rather than scaling the one that worked.

Quarter	What the Series A funds	ARR target
Q1 2029	2 more sales hires (AE + SDR). First dedicated CS hire. SOC2 audit complete.	\$600K–800K
Q2 2029	Enterprise motion starts: first 3 contracts at \$10K–15K/month. Product hardening for multi-tenant.	\$800K–1M
Q3 2029	NRR driven by expansion: existing customers grow from \$3K to \$6K/month. New logo pipeline runs without founder.	\$1M–1.4M
Q4 2029	Monthly cash-flow positive. Headcount 12–18. Founder salary at market rate (\$8K–12K/month each). No further capital required to keep the company alive.	\$1.5M–2M

Table 6.25: Series A deployment and path to self-sufficiency by end of 2029

Self-sufficient does not mean you stop raising

“Self-sufficient by end of 2029” means the company does not *need* to raise to survive. It may still choose to raise a Series B to accelerate. The difference is critical: raising from necessity means accepting any terms. Raising from strength means choosing the right partner on your own schedule.

Why Companies Like Connector Fail

This section exists because most infrastructure startups with strong technical architecture do not make it to \$1M ARR. The reasons are specific, recurring, and entirely avoidable if named honestly in advance.

- ICP sprawl.** Connector’s architecture can serve healthcare, fintech, education, and enterprise. If the team tries to sell to all four simultaneously, the product becomes generic and the sales message becomes vague. The customer says “interesting” and does not sign. The fix is to serve one ICP until you have 8 paying logos in that vertical, then expand. Connector’s 2026 ICP is narrow: a funded AI startup or mid-market SaaS company deploying LLM agents with compliance pressure. Nothing else.
- Architecture over delivery.** The Connector codebase has a Memory Kernel, a 6-pass CLS compiler, a cognitive substrate with 11 layers, and a 7-layer CNP protocol. None of these are interesting to a paying customer. What is interesting is a working audit trail for their specific agent workflow. Companies with deep architecture die when the team spends all year refining internals instead of shipping the one feature a customer is waiting for.
- Wrong first customer.** One large enterprise excited about the platform in Q1 2026 is a trap. Their procurement process will consume 8 months. They will ask for SOC2, legal review, a custom MSA, and a sandbox environment that does not exist yet. By Q3 2026 the deal is dead, the team is demoralized, and they have no paying customers and no story for investors.
- Founder cash starvation.** A founder who cannot pay rent makes three irrational decisions within six months: takes an investor’s bad terms because they need the cash,

accepts a customer's wrong requirements because they need the revenue, or burns out and walks away. This is the single most common cause of death for early infrastructure startups. It is solved in planning, not in execution.

5. **Underpricing.** Pricing at \$500/month to be “accessible” means 100 customers to reach \$600K ARR. Two founders cannot manage 100 customer relationships while building the product. Pricing at \$3,000/month means 17 customers. The temptation to underprice is strong when rejection stings. The correct answer to rejection is not lowering the price — it is finding a different customer who feels the pain more acutely.
6. **Over-engineering the demo.** A technical demo that takes 30 minutes to explain and requires the founder to be present loses 60% of potential customers before the value lands. The demo must answer one question in under 5 minutes: “what breaks for you today that this fixes?” Everything else is noise.
7. **Chasing investors before customers.** Spending months in investor meetings without paying customers produces the same answer from every credible investor: “come back when you have revenue.” The time cost is 3–4 months of lost execution. The psychological cost is a demoralized team that feels like they are failing because investors said no. Investors do not validate the business. Customers do.
8. **Single-founder technical dependency.** If only one person understands the codebase deeply and that person burns out, gets sick, or leaves, the company has a 30-day survival window. The Tech 1 hire must be someone who can eventually own the architecture independently. Documentation of internal systems is not optional — it is company-survival infrastructure.
9. **Free pilots.** Offering a free pilot “to build goodwill” achieves the opposite. It teaches customers that Connector has no pricing conviction, which signals that the product is not actually worth paying for. A customer who does not pay does not take onboarding seriously, does not give useful feedback, and cannot become a reference. Every pilot is paid from day one, even if the price is \$500/month. Price signals value.
10. **Ignoring NRR until it is too late.** Acquiring 10 pilots and losing 6 is catastrophically worse than acquiring 6 and keeping all 6. A churn rate above 20% in the first year is terminal — it means the product is not solving a real problem or the customer was wrong to begin with. NRR must be reviewed after every pilot's 3-month mark. One churned customer is a signal. Two is a pattern that requires immediate product or ICP correction.

The company can fail. This is how.

Connector fails if: (a) both founders cannot pay themselves for more than 3 months and one of them takes a full-time job, (b) the first 3 pilot attempts fail and the team pivots ICP without understanding why they failed, or (c) the product ships a critical data breach or compliance failure at a pilot customer before SOC2 is obtained. None of these are technical failures. All three are planning failures. They are named here so they can be planned against explicitly.

What Pilots Actually Get: The Value, Not The Price

The question that closes a pilot is not “how much does it cost?” It is “what do I have at the end of six months that I do not have today?” Every pilot conversation starts with the deliverable, not the price. The price is justified by the deliverable, not the other way around.

The Core Problem Connector Solves

Every AI-forward company deploying LLM agents in 2025–2026 faces a version of the same problem: *the agent makes decisions and no one can fully explain, constrain, or audit them at the infrastructure level.* Application-layer logging exists but can be bypassed, edited, or simply insufficient for a compliance auditor. The board, the compliance team, or the enterprise client asks “prove that your AI operated within policy” and the engineering team’s answer is a manual log parse that takes days and still does not produce a signed proof.

Connector installs a layer *below* the application code that:

- Intercepts every agent memory read, write, and decision at the kernel level — not at the app layer.
- Evaluates that action against an 11-layer policy chain before allowing or blocking it.
- Produces a cryptographically signed receipt of the outcome: what was decided, what data was accessed, which policy was applied, who authorised the capability, and when.
- Exposes that receipt in a format that a human auditor, a compliance tool, or a regulator can read without a developer present.

That is the product. Not “AI governance infrastructure.” Not “an agent operating system.” The product is: **a signed proof that your AI agent did what it was supposed to do — or didn’t, and here is exactly why it was stopped.**

Startup Pilot Tier: \$1,000–2,000 / month × 6 months

Target: post-seed AI startup or boutique AI consulting firm, 8–40 employees.

Deliverable	What the customer has at the end of 6 months
Audit trail	Every agent decision in their primary workflow is logged at the kernel level with a CID-addressed, signed receipt. Not application logs. Not request/response pairs. A tamper-evident proof chain.
Policy enforcement	3–5 custom policy rules authored to their specific workflow (e.g. “agent cannot access PII fields without an explicit capability token”, “agent budget cannot exceed N calls per session”). Enforced at the kernel level before the action executes.
Compliance starter	A one-page evidence export they can show to their Series A investors, their own enterprise clients, or an early SOC2 auditor: “our AI agents operate under these policies, here is the proof from the last 30 days.”
Deployment	A running Connector node in their infrastructure (Docker Compose single-node). The founders assist with setup on day one.
Integration	Their primary agent workflow (1–2 agent types) is integrated with the Connector kernel. SDK and worked examples provided.
Roadmap access	Direct Slack channel. Feature requests logged and prioritised. Response within business-hours on the same day.
Early pilot exclusive	Price locked for 18 months post-pilot. Named as a design partner on the website (if agreed). Co-authorship of one integration pattern that ships in the official SDK. First access to the CLS authoring interface when it launches.

Table 6.26: Startup pilot tier — deliverables for \$1,000–2,000 / month over 6 months

The one-sentence pitch for this tier: *“Your board asked what happens when your AI makes the wrong decision. After six months with Connector, your answer is a signed receipt showing exactly what it decided, why, and which policy it was operating under — provable without a developer in the room.”*

Mid-Enterprise Pilot Tier: \$5,000–7,000 / month × 6 months

Target: Series A or Series B company, 40–200 employees, with 2–5 internal AI agent workflows in production, compliance team already asking questions.

Deliverable	What the customer has at the end of 6 months
Full multi-workflow coverage	All 2–5 of their production agent workflows instrumented. Not just one. Each workflow has its own policy set, its own receipt chain, and its own compliance export.
Compliance report export	Monthly one-click export: a PDF or CSV that their compliance team or external auditor accepts as evidence. Fields covered: agent identity, action taken, policy applied, data accessed, outcome (allowed / blocked), timestamp, cryptographic signature. Specifically structured for HIPAA or SOC2 CC6/CC7 evidence requirements.
Custom policy authoring	8–12 policy rules authored together with the customer’s compliance or engineering team. These are not template rules — they are written for their specific data types, regulatory context, and agent behaviour patterns.
Real-time policy violation alerts	When an agent action is blocked by the kernel, an alert fires to their incident management system (PagerDuty, Slack, webhook). The alert includes: which agent, which rule, what it attempted, and what data it was trying to access.
Dedicated onboarding	Founders spend 2–3 days on-site or in intensive remote sessions to complete integration. Not a self-service process.
SLA for response	4-hour business-hours response on any production issue via dedicated Slack channel. Founders are directly contactable.
Quarterly review	A 60-minute call each quarter reviewing agent behaviour trends, policy violation patterns, and recommended policy improvements.
Early pilot exclusive	Annual contract option at 15% discount (\$51K–\$71.4K/year vs. \$60K–\$84K at list price). Co-authored case study (published anonymously if required). Input rights on the enterprise roadmap for the following 12 months. First access to multi-tenant and multi-node features before general availability.

Table 6.27: Mid-enterprise pilot tier — deliverables for \$5,000–7,000 / month over 6 months

The one-sentence pitch for this tier: *“You have five AI agent workflows in production. Your compliance team asks quarterly: prove they are operating within policy. After six months with Connector, that question is answered with a one-click export — every agent action, every policy check, every data access, signed cryptographically — structured specifically for your HIPAA or SOC2 audit requirements. And when an agent breaks a rule, the kernel stops it before the action happens and alerts your on-call team within seconds.”*

What Early Pilots Get That Later Customers Never Will

The first 10 pilot customers occupy a position that no subsequent customer can buy:

1. **18-month price lock.** When Connector raises Seed 1 in Q4 2027 and reprices to reflect the product's maturity, early pilots keep their pilot rate for 18 months from signing. Given the trajectory, this is likely a 30–50% discount against future list price.
2. **Direct founder access, permanently.** After Seed 1, the founders will be managing a team and a sales pipeline. Early pilot customers retain a Slack channel directly to the founders for the duration of their subscription. This is not offered to post-Seed customers.
3. **Roadmap influence.** The first 10 pilots collectively shape 30–40% of the 2027 product roadmap. They are consulted before features are scoped, not after they are shipped.
4. **Reference designation.** Customers who agree to be a named reference become part of the sales story for Seed 1. Their credibility raises capital at better terms, which means the company is more likely to survive and serve them well long-term. Being a reference is a direct investment in the vendor's success.
5. **Co-authorship in the SDK.** The integration patterns the first 10 pilots use become the official SDK examples. Their workflow is the one every new customer sees first. That is structural product influence that persists beyond their contract.
6. **Pre-announcement of enterprise features.** Before any new feature ships, early pilot customers are notified privately and given 2 weeks to test and provide feedback before general availability. They always have at least one version-cycle advantage over the general customer base.

The pilot is not a discount. It is a co-investment.

An early pilot customer is not getting a cheaper version of a finished product. They are co-investing with their time, feedback, and money in building a product that is shaped around their use case. The price reflects that relationship. A customer who understands this signs. A customer who thinks they are buying discounted enterprise software will be disappointed by the rough edges and will churn. Qualify this distinction in every first conversation.

The Real Plan: Zero to \$30K MRR, Month by Month

This section is not a projection. It is an operating plan for two specific founders, one of whom has real financial pressure right now, in a codebase that is architecturally strong but not yet customer-ready. Every week matters. Every decision either extends runway or consumes it.

The honest starting condition — March 2026

One founder needs a salary and has real-life financial obligations that cannot wait for a product-market fit story. This is not unusual and not shameful. It is a constraint that must be treated the same way as a technical constraint: acknowledged, planned around, and resolved as fast as possible. The plan below is built around this constraint, not despite it. The fastest path to a salary that covers real life is two paying pilots by end of July. Everything else is in service of that goal.

Before May: What Must Happen Right Now

The window from now until the end of April is the only period where the founders still have some breathing room. Use it for two things only:

1. **Generate one consulting income event.** One short-term project — 2–3 weeks, \$3,000–5,000 — extends personal runway by 6–10 weeks. One founder takes this. The other builds the demo. The goal is to arrive at May with \$8,000–12,000 in combined pocket rather than \$0–5,000. This is not ideal but it is the difference between having a runway and not having one. Do not wait. Identify the project this week.
2. **Complete the demo and deployment.** Docker Compose single-node. Works in one command. A script that walks through one governed agent workflow in under 10 minutes. Three worked examples in the integration doc. This is the entire Phase 0 deliverable. Nothing else gets built until this works reliably. Two founders, four weeks, full focus.

Month-by-Month: May 2026 to November 2026

Month	State	What happens	MRR	Founder salary
May	\$0–5K pocket	Demo is ready. Outreach list of 75 CTOs built. Send first 40 messages. Target: 6–10 responses, 3–5 demos scheduled. One founder on consulting 1 day/week if needed. No revenue yet.	\$0	\$0 from company
June	Demos running	Run 8–12 demos. Qualify hard. Three serious conversations. Two LOI drafts in progress. Do not drop price. If a prospect says “we like it but \$3,000 is too much” that is the	\$0–\$3K (1–2 early invoices)	\$0–\$1.5K

The Personal Salary Resolution — Month by Month

This is the part that is usually not written down. It needs to be.

Month	MRR	What the founder can realistically take	Gap to cover
March–April	\$0	\$0 from company. Consulting income if arranged. Draw on savings.	\$3,500–5,000/mo personal
May–June	\$0–\$3K	\$0–\$1,500 from company. Still need savings or consulting. Tight.	\$2,000–5,000/mo
July	\$6K–\$8K	Company pays one founder \$3,000–\$3,500/month. The other either defers or uses savings for 4–6 more weeks.	\$0–\$2,000/mo per person
August	\$10K–\$14K	Both founders take \$4,000/month. Still lean but survivable in most cities.	\$0–\$1,000/mo
September	\$14K–\$20K	Both founders take \$5,000/month. Covers rent, food, basics. Not comfortable but stable.	Covered
November	\$25K–\$35K	Both founders take \$6,000/month. Real life bills are covered. Financial stress drops.	Covered with buffer
Q1 2027	\$35K–\$50K	Post pre-seed close, both founders formalize \$6,500–7,500/month salary. First time it feels real.	Fully covered

Table 6.29: Personal salary resolution timeline — honest month-by-month

July is the minimum viable month for personal survival

If pilots 1 and 2 are not signed and invoiced by end of July, the personal financial situation becomes a company crisis. One founder will be forced to take outside income full-time. This is not failure — it is a signal that the ICP or the demo needs to change, not that the company is dead. If July closes without revenue: spend two weeks understanding why the demos did not convert, fix exactly one thing, and restart outreach. Do not pivot the entire product. Fix the conversion, not the product.

What “Design Partner” Means in Practice

Two design partners in July–August 2026 does not mean two customers who use the platform occasionally. It means:

- A contract signed and a card charged before any integration work begins. Even \$1,000 upfront. Money before work.
- A weekly 30-minute call, on a fixed day, every week for the first 8 weeks. The founder drives the agenda. The customer shows up because they paid.
- A specific workflow already identified before the first call: “your agent that does X — that is what we are going to govern first.” Not a general deployment. One workflow, done well.
- A clear success criterion agreed in writing at signing: “at the end of 6 months, you will have a signed audit trail for every decision this agent makes, exportable to a one-page report for your compliance team.” Not vague. Specific and measurable.
- The founders treat this customer’s bugs as production incidents. Response within 2 hours. A fix within 24 hours or a workaround with an ETA. This is not normal startup product behaviour — it is the reason they pay \$3,000–\$4,000/month for something that is not fully finished.

2027: Grounded Quarterly Plan

Entering 2027, Connector has 7–10 paying pilots, \$25K–35K MRR, a working product, and a real customer story. Q1 2027 is still a single-track execution: close the pre-seed, bring Tech 1 in, renew the 2026 pilot cohort. From Q2 2027 the business runs on **three commercial tracks simultaneously**. These tracks are not alternatives — they are parallel and permanent.

Three Commercial Tracks (Q2 2027 onwards — all three run forever)

The three tracks map directly to the licensing tier model already implemented in code: `Tier::Growth/Business` for Track 1, `Tier::Indie/Startup` for Track 2, and `Tier::Enterprise/Core` (with `features_override: ["hipaa", "soc2"]`) for Track 3.

#	Track	Example	Why this track can not be abandoned	Tier (code)
1	Survival Bridge	Mid-startup & small business	Economic growth is the support revenue SaaS floor. Even if startups, every tech entrepreneur — the deal slips com-by pa- 6 nies months, from these the cus- 2026 ICP keep who the re- lights new on and and new salaries com-paid. pa- Never nies de- of pri- the ori- same rise type. this 20- track 50 re- em- gard- ploy-less ees. of AI en- in ter- pro- prise duc- pipeline tion. size. Monthly pain.	Business \$1,000/mo

Track 1 is the survival constraint, not a stepping stone

Connector's real long-term customers are enterprise in regulated segments. But enterprise deals slip. A 3-month slip in one enterprise close is normal. A 6-month slip happens regularly. If the company has abandoned Track 1 to focus on enterprise, a slip means payroll stress and founder salary cuts. Track 1 must always have active outreach and active closes running in parallel with the enterprise motion, unconditionally. This is not optional in good times and it is the only thing that matters in bad times.

Quarterly Plan with Three-Track Breakdown

Quarter	MRR target	Operational focus	Key event
Q1 2027	\$35K–\$50K	<i>Single track.</i> Close pre-seed SAFE (\$75K–\$125K). Tech 1 joins. Renew 2026 pilot cohort to annual contracts. Price for new Track 1 customers rises to \$800–\$1,200/month. Track 2 self-serve portal goes live. No enterprise outreach yet — build the compliance feature flag pipeline (<code>hipaa</code> , <code>soc2</code>) so Track 3 is ready.	Pre-seed closes
Q2 2027	\$50K–\$65K	<i>Three tracks start.</i> Track 1: continue ecommerce/agent outreach, target 5 new Track 1 closes. Track 2: self-serve portal active, 30–50 small business sign-ups at Indie/Startup tier, no founder time needed per customer. Track 3: first 2–3 enterprise pilot conversations start (healthcare or fintech). Use <code>PilotGrant</code> with <code>tier_override = Enterprise</code> and features = [<code>hipaa/soc2</code>]. Seed investor shortlist of 10–15 names built.	3-track begins; first enterprise pilots
Q3 2027	\$65K–\$85K	Track 1: 25–30 active mid-startup customers at \$800–\$1K avg = \$20–\$30K. Track 2: 60–80 self-serve customers at \$200 avg = \$12–\$16K. Track 3: 3–4 enterprise pilots paying = \$9K–\$16K. Formal Seed 1 fundraising begins. Tech 1 carries product. Founders split 60% / 40% execution vs. fundraising. Track 1 must not stall during	Seed 1 process begins

What Makes 2027 Work or Fail

The 2027 plan hinges on three things that cannot be bought or rushed:

1. **Annual renewal rate from 2026 pilots.** If fewer than 60% of the November 2026 pilots renew into 2027, the Seed story is broken. The product is not retaining. The fix is not more sales — it is diagnosing why the first cohort did not renew and fixing it before adding more customers. One churned pilot investigated honestly is worth more than five new ones chased blindly.
2. **Tech 1 hire quality.** Tech 1 joins in Q1 2027. If this person cannot independently handle customer integration issues within 6 weeks of starting, the founders cannot give 40% of their time to fundraising in Q3. The Seed process dies because product execution stalls. Hire slowly. One wrong Tech 1 sets the company back 4 months.
3. **Founder bandwidth during Seed fundraise.** Raising Seed takes 60–90 hours of active founder time per month for 2–3 months. Product decisions slow down. Customer response times increase. This is unavoidable. The mitigation is: (a) give existing customers a heads-up that the founding team is in a fundraise period, (b) ensure Tech 1 has a documented escalation path for production issues, and (c) do not start the fundraise until the product is stable enough to run for 4 weeks without a founder intervention.

The only metric that matters at the end of 2027

At the end of Q4 2027, one number determines whether the company is on track: the 12-month NRR of the 2026 pilot cohort. If it is above 110% — meaning pilots renewed and some expanded their contract — the Series A conversation in 2028 is credible. If it is below 90%, no amount of new logos fixes the story. Protect the 2026 cohort as if they are the only customers you will ever have, because in 2027, they effectively are.

Honest Numbers: What Self-Sufficiency Actually Looks Like at Each Stage

Period	MRR	Team burn	Monthly net
Nov 2026	\$30K	2 founders at \$6K = \$12K total	\$+18K (reinvested in hiring pipeline)
Q1 2027	\$42K	2 founders + Tech 1 = \$20K	\$+22K
Q2 2027	\$58K	+ Tech 2 = \$26K	\$+32K
Q4 2027	\$88K	+ Sales 1, formalised salaries = \$40K	\$+48K (strong Seed 1 narrative)
Q4 2028	\$140K	8–10 people at \$80K–\$100K total	\$+40K–\$60K
Q4 2029	\$160K–\$185K	12–16 people at \$130K–\$150K	Cash-flow positive. No survival dependency on next round.

Table 6.32: Team burn vs. MRR from November 2026 through end of 2029

This plan requires that July 2026 actually closes revenue

Every number in 2027 and 2028 is downstream of two pilots signing and paying in July 2026. If that does not happen, the cascade shifts by one quarter: August closes, October hits \$7K, January 2027 hits \$30K. That is survivable. What is not survivable is not closing the first pilot before October 2026. After that date, the founder's personal financial situation forces a decision that may end the company. The entire strategic plan lives or dies in a 30-day window in July. Treat it accordingly.

Chapter 7

Node Operations

7.1 The Node as Infrastructure

A Connector node is not just a process — it is a governed runtime that manages agents, memory, and protocol sessions with the same rigour as a database server or a message broker. This chapter covers what happens from the moment `connector-node` starts to the moment it shuts down gracefully, and every operational concern in between.

7.2 The 12-Stage Boot Sequence

Connector nodes boot through a deterministic 12-stage pipeline. Each stage either succeeds and advances to the next, or fails and halts the boot with a clear diagnostic. There is no partial initialisation and no silent degraded mode — the node is either fully ready or it does not serve traffic.

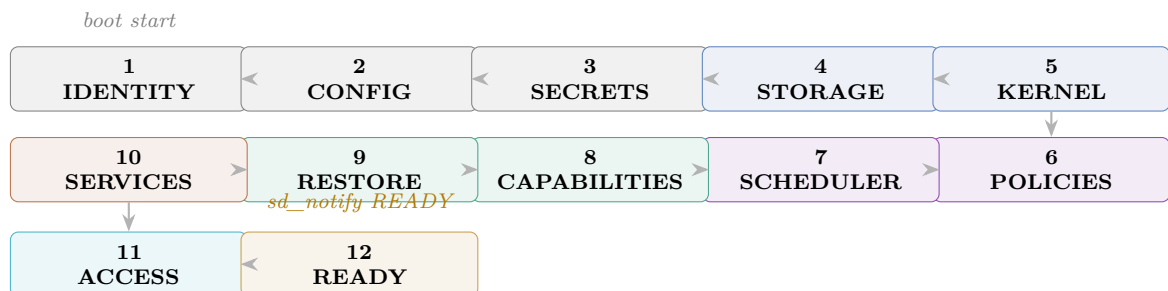


Figure 7.1: Connector node 12-stage boot pipeline — any stage failure halts the boot

#	Stage	What happens
1	IDENTITY	Load or generate node DID; load Ed25519 signing keys
2	CONFIG	Parse <code>config.toml</code> , env vars, CLI flags; resolve hierarchy
3	SECRETS	Load API tokens, LLM keys, database credentials from vault/file
4	STORAGE	Connect to storage backend (PostgreSQL/SQLite); verify schema
5	KERNEL	Boot Memory Kernel; verify CID chain integrity; load namespace index
6	POLICIES	Load governance predicates and the 11-layer predicate chain
7	SCHEDULER	Initialise agent scheduler; allocate resource pools
8	CAPABILITIES	Register tool providers: built-in, MCP servers, CNP ports
9	RESTORE	Reload agents and sessions from last checkpoint; resume execution
10	SERVICES	Start HTTP API, WebSocket, CNP router, A2A and MCP bridges
11	ACCESS	Enable local dashboard; enable API key authentication
12	READY	Send <code>sd_notify READY</code> ; start health probe endpoints

Table 7.1: 12-stage boot — description of each stage

Each stage writes a boot-progress log line with its name, status, and elapsed time. The terminal output follows a fixed format: `[N/12] STAGE-NAME OK (0.42s)`, which makes boot logs scannable by eye and parseable by automation.

7.3 systemd Integration

Connector nodes are designed as systemd services. The key integration points are:

- **Type=notify** — systemd waits for `READY=1` before marking the service as started, preventing premature traffic routing.
- **WatchdogSec=60s** — the node must send a watchdog heartbeat every 60 seconds or systemd kills and restarts it.
- **Restart=on-failure** — automatic restart on any non-zero exit; agents resume from checkpoint on next boot.

7.4 Health Probes

Connector exposes three health probe endpoints compatible with both Kubernetes and traditional load balancers:

Probe	Endpoint	What it checks	Failure means
Liveness	GET /health/live	Node event loop is not dead-locked	Kill and restart immediately
Readiness	GET /health/ready	Kernel scheduler, and storage are all operational	Remove from load balancer; stop sending traffic
Startup	GET /health/startup	All 12 boot stages have completed	Do not start other probes yet; give more boot time

Table 7.2: Health probe endpoints — Kubernetes and load-balancer compatible

Probe ordering in Kubernetes

Configure the startup probe first with a generous **failureThreshold** (e.g., 30 attempts $\times$ 10s = 5 minutes). Only after startup succeeds do the liveness and readiness probes activate. This prevents Kubernetes from killing a slow-starting node during a large checkpoint restore.

7.5 Agent Deployment

Agents are deployed from a TOML manifest file. The manifest declares the agent’s identity, runtime requirements, tool capabilities, governance predicates, and resource budget.

7.5.1 Deployment Flow

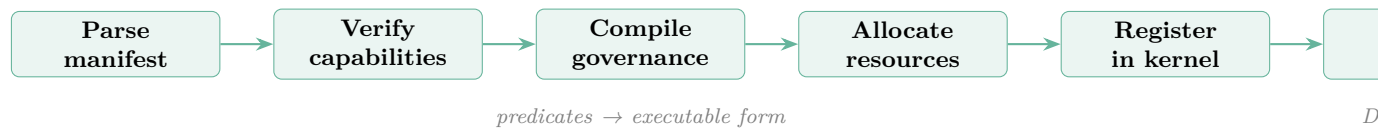


Figure 7.2: Agent deployment pipeline — manifest to running agent

7.5.2 Agent Manifest Structure

The manifest is a typed, validated TOML document. Four top-level sections define the agent:

Section	What it declares
[agent]	Name, description, version, DID prefix
[runtime]	LLM model, temperature, max tokens, cognitive substrate config
[capabilities]	Tool names, memory namespaces, protocol ports to open
[governance]	Predicate names, resource budget (tokens, memory, time, calls)

Table 7.3: Agent manifest sections

7.6 Monitoring and Observability

7.6.1 Prometheus Metrics

The node exposes a `/metrics` endpoint in Prometheus text format. Key metrics exported:

Metric name	Type and description
<code>connector_agents_running</code>	Gauge — active agent count
<code>connector_sessions_total</code>	Counter — sessions opened since start
<code>connector_memory_packets</code>	Counter — MemPackets written
<code>connector_tokens_consumed</code>	Counter — total LLM tokens used
<code>connector_predicate_checks</code>	Counter — predicate enforcement calls
<code>connector_predicate_denials</code>	Counter — predicate enforcement denials
<code>connector_tool_calls_total</code>	Counter — external tool invocations
<code>connector_boot_duration_ms</code>	Gauge — last boot time in milliseconds
<code>connector_receipts_issued</code>	Counter — signed execution receipts

Table 7.4: Core Prometheus metrics exposed by the node

7.6.2 Structured Logging

All node logs are emitted in structured JSON via the `tracing` framework. Every log line carries a consistent set of fields: `timestamp`, `level`, `target` (module), `agent_did`,

`session_id`, and `message`. This makes logs directly ingestible by Loki, Elasticsearch, Splunk, or any JSON-aware log pipeline.

7.6.3 OpenTelemetry Tracing

Long-running operations (agent execution, contract compilation, checkpoint restore) emit OpenTelemetry spans. Each span carries the agent DID, session CID, and predicate enforcement result. Traces can be exported to Jaeger, Tempo, or any OTLP-compatible collector.

7.7 Graceful Shutdown

Connector nodes shut down in a deterministic eight-phase sequence on receipt of `SIGTERM` or `SIGINT`. No agent is killed mid-execution; all in-progress sessions are checkpointed before the process exits.

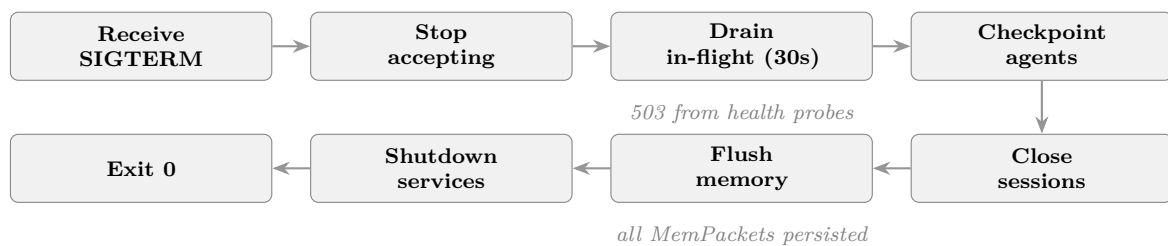


Figure 7.3: Graceful shutdown sequence — 8 phases, no data loss

7.8 Backup and Restore

The kernel’s content-addressed store makes backup straightforward: a backup is a consistent snapshot of the MemPacket store. Because every packet has a CID, incremental backups are simple set differences.

```

1 connectorctl backup memory --output /backups/$(date +%Y%m%d).tar.gz
2 connectorctl backup memory --incremental --since 1h --output /backups/inc.
  tar.gz
3 connectorctl restore memory --input /backups/20260322.tar.gz
  
```

Agent-level backups capture the agent’s full MemPacket namespace and the last signed checkpoint, making it possible to migrate an agent to a new node with full history intact.

7.9 Disaster Recovery

Scenario	Recovery procedure	Data loss
Node crash	Restart node; agents resume from last checkpoint on stage 9	None
Disk corruption	Restore from backup; replay audit receipt chain	Up to last backup
Database failure	Switch to replica (if configured); restore from WAL	Seconds
Network partition	Agents pause; session receipts held in local queue; resume on reconnect	None
License expiry	Agents pause gracefully; resume after license renewal via <code>connectorctl activate</code>	None
Kernel integrity fail	Stage 5 halts boot; operator restores from backup or runs <code>connectorctl repair</code>	None

Table 7.5: Disaster recovery scenarios and procedures

Metric	Single-node	Cluster (3+ nodes)
RTO (Recovery Time Objective)	≤ 5 minutes	≤ 30 seconds
RPO (Recovery Point Objective)	Last checkpoint (<1 min)	Near-zero (replica sync)

Table 7.6: RTO and RPO targets by deployment topology

7.10 Deployment Topologies

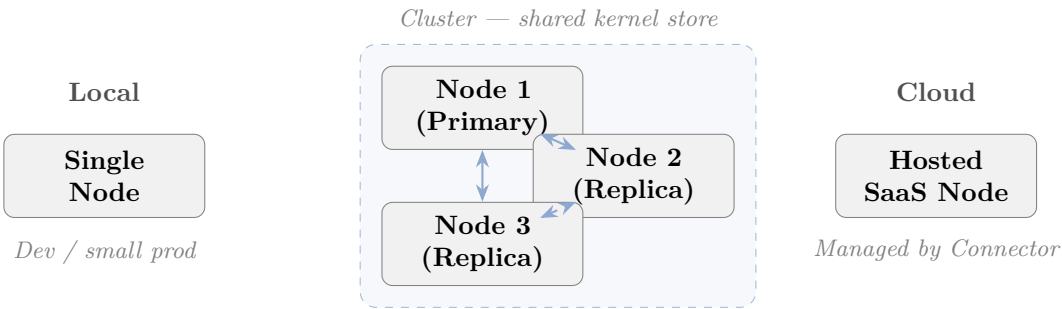


Figure 7.4: Supported deployment topologies — single-node, cluster, and hosted SaaS

	Single-node	Cluster	Hosted SaaS
License tier	Free–Enterprise	Enterprise	Any
HA	No	Yes	Yes
Data location	Customer premises	Customer premises	Connector-managed
Management	Self-managed	Self-managed	Fully managed
Best for	Dev, small prod	Large prod, HA	No ops team

Table 7.7: Deployment topology comparison

Chapter 8

Growth Trajectory: 2027–2029 and Beyond

8.1 Why 2026 Execution Defines the Entire Trajectory

The relationship between 2026 and every year that follows is not linear — it is compounding. A company that closes ten paid pilots with pricing discipline and at least four annual conversions by the end of 2026 does not just have revenue. It has **proof of wedge, proof of repeatability, and proof that enterprise buyers in at least one vertical are willing to pay for governed AI.**

The compounding logic

Investors do not fund metrics. They fund the story those metrics tell. Ten paid pilots, \$200k–\$400k ARR, a buyer map across two or three regulated verticals, and a clean conversion rate from pilot to annual tells a very specific story: *“This team found the wedge, sold it at price, and kept the customer.”* That story funds a Seed round. The Seed round funds the team that 10x the story. The 10x story funds Series A.

If 2026 targets are missed — fewer pilots, no pricing discipline, poor retention — the trajectory compresses toward one of two paths: a hard pivot or a slow wind-down. This chapter assumes the 2026 plan in Chapter 6 is executed with discipline.

8.2 2027: Expansion and Seed 1

8.2.1 Starting position

A realistic 2026 exit state that supports a Seed 1 raise in early-to-mid 2027 looks like this:

Metric	Target by end of 2026
Paid pilots closed	8–10 (healthcare, fintech, enterprise platform)
Converted to annual	3–5 contracts, \$15k–\$30k ARR each
Total ARR	\$80k–\$150k (early, but directionally clean)
Gross logo retention	Near 100% in pilot year
Funnel depth	20–30 active conversations, 10–15 in late-stage
Reference customers	2–3 willing to speak to investors
Team	2–4 people (founder-led, lean)

Table 8.1: 2026 exit state that supports a credible Seed 1 raise

8.2.2 What a Seed 1 raise looks like in 2027

The 2025–2026 market context matters: Founder Institute’s 2025 benchmark data showed Seed rounds concentrating toward smaller numbers of higher-quality bets, with investors expecting paid traction even at this stage. For Connector, the Seed 1 raise is therefore **a validation round, not an exploration round**.

Raise parameter	Realistic estimate
Raise size	\$1.5M–\$3.5M
Pre-money valuation	\$8M–\$15M
Equity dilution	15–25%
Target investors	Infrastructure-focused angels, pre-seed/Seed funds, enterprise AI specialists
Lead signal	Paid ARR + clear wedge + 2 strong reference customers + founder-led GTM evidence
Timeline	Q1–Q2 2027 (close); requires 4–6 months of raise process

Table 8.2: Realistic Seed 1 raise parameters for Connector in 2027

8.2.3 What Seed 1 capital funds

- **Engineering hires (2–3 people):** multi-node support, enterprise integrations (SSO, SCIM, SIEM), Glue CLI maturity, compliance tooling (HIPAA, SOC2 evidence pipeline).
- **Customer success / implementation (1 person):** reduce pilot launch time, increase conversion rate, build repeatable onboarding playbook.
- **Commercial hire or fractional (1 person):** support founder in closing mid-market and enterprise deals in two target verticals.
- **Compliance and trust assets:** security review, pen test, privacy documentation — the artefacts that unblock procurement in healthcare and finance.

8.2.4 2027 commercial targets

With Seed 1 capital and a structured team:

Metric	Target by end of 2027	Honest range
ARR	\$500k–\$1M	\$300k floor, \$1.5M ceiling
Paying customers	25–40	Depends on conversion rate from 2026 pipeline
Average contract value	\$15k–\$25k ARR	Unlikely to exceed \$40k without enterprise motion
Monthly burn	\$80k–\$120k	Post-hire run rate
Months of runway	18–24	At raise size and burn
Gross NRR	>110%	Expansion from pilot to larger contracts
MRR growth	8–12% month-on-month	If GTM is working

Table 8.3: 2027 commercial targets and realistic ranges

8.2.5 2027 business SWOT

Dimension	Honest 2027 assessment
Strengths	Proven wedge across two verticals, first paying customer references, small team with high output, data-sovereignty architecture differentiates from hosted competitors
Weaknesses	Still founder-led sales (not yet scalable), product missing enterprise-grade integrations, brand awareness near zero, multi-node still early, category education burden still high
Opportunities	Governance regulation accelerating globally (EU AI Act enforcement, US EO on AI), large incumbents slow to add runtime-native audit, bottom-up adoption by AI engineers creates organic pipeline
Threats	Larger infra vendors (Datadog, Snowflake) may add partial observability/-governance features, a well-funded competitor could outspend on GTM, economic headwinds may freeze enterprise AI budgets, team attrition at small company is high-risk

Table 8.4: Business SWOT for 2027 expansion stage

8.2.6 Real risks in 2027 — not a sanitised list

The biggest risks in 2027 are not market risks. They are execution risks.

Honest risk register for 2027

- **Founder bottleneck.** If every deal still runs through the founder, the company cannot scale past 20 customers without burning the team. Fix: hire CS and fractional sales early.
- **ICP drift.** The 2026 pilots are concentrated; 2027 capital creates pressure to expand too fast across too many verticals. Fix: stay in two verticals until unit economics are confirmed.

- **Product sprawl.** New hires want to build features. Features without customer traction create maintenance debt that slows the core wedge. Fix: product backlog must be driven by sales blockers only.
- **Enterprise compliance lag.** Healthcare and fintech buyers may stall on procurement until SOC2 or HIPAA BAA evidence is available. Fix: start compliance programme in Q1 2027 with Seed capital.

8.3 2028: Category Definition and Seed 2 / Pre-Series A

8.3.1 Where Connector should be by end of 2027

For 2028 to be a category-definition year, the company must enter it with:

- ARR in the range of \$500k–\$1M with clear expansion mechanics.
- A repeatable sales process that does not require the founder in every deal.
- Multi-node and cluster deployments working in at least two enterprise accounts.
- A documented compliance posture (SOC2 Type I minimum).
- A clear opinion on what Connector is: not just the debugger and not yet the entire AI runtime stack, but the **governance and evidence control plane** that enterprises insert between their AI teams and production.

8.3.2 The 2028 market context

By 2028, the enterprise AI governance market will be materially larger and more structured. The EU AI Act enforcement timeline, US regulatory activity, and the natural maturation of agentic AI deployments will mean that governance is no longer a nice-to-have but a procurement requirement in regulated industries.

The Gartner AI Governance Platforms category, which was still forming in 2025, will likely have 15–20 named vendors by 2028 with more rigorous analyst coverage. This is simultaneously an opportunity (the category exists and has budget) and a threat (more direct competitors with funding).

8.3.3 Raise options in 2028

At \$500k–\$1M ARR with strong retention and multi-vertical evidence, Connector has two credible paths:

Path	When it makes sense	Parameters
Seed 2 (bridge)	ARR is \$500k, growth is strong but Series A bar not yet met	\$2M–\$4M; extends runway 18 months; funds scale to Series A gate
Series A	ARR is \$800k+, growth rate >100% YoY, 2+ verticals with expansion, clear GTM machine	\$6M–\$12M; funds engineering scale and enterprise sales motion

Table 8.5: 2028 raise paths depending on execution velocity

The Series A gate is well-documented

Y Combinator’s 2025 guidance and Lightspeed’s 2024 benchmark data both show that B2B SaaS Series A typically requires: \$1M+ ARR growing faster than 2x YoY, evidence of an efficient GTM motion (CAC payback under 18 months), and a clear category claim. For Connector, the harder bar is the **GTM efficiency requirement**, not the ARR. Founder-led sales alone will not satisfy that gate.

8.3.4 2028 commercial targets

Metric	Target by end of 2028	Honest range
ARR	\$1.5M–\$3M	\$800k floor; \$4M ceiling with strong PLG
Paying customers	60–100	Includes both self-serve Pro and enterprise
ACV	\$20k–\$40k	Enterprise deals at \$40k–\$80k possible
Team	12–20 people	Engineering, CS, sales, compliance
Burn rate	\$200k–\$350k/month	Post-Series A or large Seed 2
Gross NRR	>120%	Land-and-expand must be demonstrably working

Table 8.6: 2028 commercial targets

8.3.5 2028 business SWOT

Dimension	Honest 2028 assessment
Strengths	Named category position, reference customers in 2 regulated verticals, SOC2 posture, repeatable sales motion starting to show, data-sovereignty moat difficult for hosted-only competitors to replicate
Weaknesses	Team size still limits sales surface, product still behind on enterprise table-stakes (complex SSO, SAML, RBAC at team level), category education still required for mid-market buyers
Opportunities	EU AI Act creates direct procurement requirement in European market, large enterprise platform teams standardising agent governance creates platform deals (\$100k+ ACV), OEM / partnership routes open with HRMS or EHR vendors
Threats	Well-funded competitor enters with PLG motion and lower price point, Datadog or Elastic launches observability + governance bundle, Series A pressure to grow faster than product quality can support

Table 8.7: Business SWOT for 2028 category-definition stage

8.3.6 Real risks in 2028

Honest risk register for 2028

- **The ARR trap.** Growing from \$500k to \$1.5M requires 3x growth in 12 months. If the sales motion is still primarily founder-led, this is very hard. The first sales hire must succeed and must happen in H1 2028 at the latest.
- **Category crowding.** By 2028, there will be more competitors. A company that has not earned a clear category position (“the governed runtime for agentic AI”) will be compared on feature checklists with well-funded alternatives.
- **Enterprise expectation gap.** Enterprise buyers who move from pilot to \$40k+ ARR contracts will demand SLAs, support response times, and integrations that a 15-person team cannot easily sustain. Fix: invest in CS and tooling, not just sales.
- **Seed 2 terms creep.** If Connector raises a Seed 2 at a flat or down valuation due to missing targets, the cap table and founder motivation can deteriorate quickly. Fix: manage cash carefully; do not over-hire in 2027.

8.4 2029 and Beyond: Stabilisation and Market Position

8.4.1 What stabilisation actually means

The word “stabilisation” is often misread as slowing down. In this context it means something specific: **the company transitions from proving it can sell to proving it can retain, expand, and operate at scale with manageable burn.** This is the transition from growth mode to efficient growth mode.

Historical SaaS data from SaaS Capital and OpenView shows that companies that reach \$3M ARR with NRR above 115% and a CAC payback period under 24 months have a very high probability of reaching \$10M ARR. The hard question is whether Connector can maintain NRR above 115% as it scales from 30 to 100 customers with a larger and more heterogeneous customer base.

8.4.2 2029+ targets and realistic scenarios

Metric	Conservative	Realistic	Optimistic
ARR by end of 2029	\$3M	\$6M–\$8M	\$10M–\$15M
Customers	80–120	150–250	300+
ACV	\$25k	\$30k–\$40k	\$40k–\$60k
NRR	105%	115–125%	130%+
Team	20–25	30–40	50+
Fundraising state	Still on Seed 2 run-way	Series A closed	Series B possible

Table 8.8: 2029 scenario analysis — conservative, realistic, and optimistic

8.4.3 What determines which scenario materialises

The difference between the conservative and realistic scenarios is not product quality. By 2029, the product will be mature enough in all three scenarios. The differentiating variables are:

- 1. **Sales hire timing and quality.** Companies that get their first true enterprise AE right in 2027–2028 compound their growth. Companies that delay or hire wrong fall into the \$1M–\$2M ARR plateau that kills many B2B infra companies.
- 2. **Category clarity.** If Connector is still explaining what it is in 2029, it will be stuck in founder-led sales forever. The category must be claimed and repeated until buyers come with the vocabulary.
- 3. **NRR.** Expansion within accounts is the multiplier. If customers expand from one workflow to one business unit to company-wide, \$3M ARR can become \$6M ARR with no new customer acquisition.
- 4. **Regulatory tailwind.** The EU AI Act, US NIST AI RMF, and similar frameworks will create procurement requirements. Companies already in procurement at regulated enterprises in 2027 will be entrenched by 2029.

8.4.4 Three exit paths from 2029 onwards

Path	Conditions	Timeline
Series B + independent scale	\$6M+ ARR, NRR >120%, clear category position	2029–2030; target \$20M–\$40M raise
Strategic acquisition	Strong vertical depth in healthcare or fintech; enterprise customer base attractive to platform buyer (Salesforce, ServiceNow, SAP, Databricks)	Possible from 2028; likely 2030–2031
Organic profitability	If growth moderates and team stays lean; reach free cash flow positive at \$5M+ ARR	Possible by 2030 if burn is controlled

Table 8.9: Three realistic exit paths from 2029 onwards

8.4.5 2029+ business SWOT

Dimension	Honest 2029 assessment
Strengths	Multi-year customer relationships in regulated verticals, deep product moat (memory kernel, signed receipts, governance runtime are hard to replicate quickly), regulatory tailwinds have created a real budget line for AI governance
Weaknesses	Still a small player in a growing but fragmented category, enterprise expectations are high and support costs scale with customer count, international expansion requires compliance per jurisdiction
Opportunities	Platform vendors (EHR, CRM, HRMS, data warehouse) looking to embed AI governance as a feature could create OEM or integration channels; open-source community edition could generate bottom-up pipeline at scale
Threats	Microsoft, Google, and AWS have distribution advantages and may bundle governance into their AI platforms; regulatory compliance may be commoditised by large vendors making it harder to differentiate

Table 8.10: Business SWOT for 2029+ stabilisation stage

8.5 The Full Arc: Three-Year Business Evolution Summary



Figure 8.1: Three-stage business arc — wedge validation, expansion, and category definition

8.6 The One Thing That Kills Each Stage

Most companies do not fail because the market does not exist. They fail because they make one avoidable mistake at each stage transition:

Stage	Most common kill shot	How to avoid it
2026	Spreading pilots across too many segments with no pricing discipline	Pick two verticals, hold price, accept fewer pilots
2027	Hiring the wrong first sales rep and keeping them too long	Define a clear ICP and quota before hiring; manage out inside 90 days if it is not working
2028	Growing revenue without earning NRR (churning customers to replace them with new ones)	Measure and report NRR monthly; do not close a deal you cannot retain
2029	Category dilution — becoming “an AI tool” rather than the governance and evidence layer	Repeat the category claim publicly and consistently; own one anchor sentence

Table 8.11: Stage-specific failure modes and mitigations

Chapter 9

Technology Evolution: 2027–2029 and Beyond

9.1 Starting State: What the Technology Looks Like After 2026

If 2026 delivers ten paid pilots and a working commercial motion, the technology that produced that result will have specific properties and specific gaps. It is important to be precise about both, because the 2027–2029 investment decisions follow directly from an honest assessment of the current architecture.

9.1.1 What works well by end of 2026

Component	Mature and working
Memory Kernel (MemPacket, CID, predicate chain)	Core abstraction is solid; content-addressed storage and signed receipts are production-ready
Single-node boot (12-stage pipeline)	Deterministic, systemd-integrated, health probes working
Agent lifecycle management	Create, start, stop, pause, resume, kill — all covered
Basic policy enforcement	Predicate chain with 11 enforcement layers operational
CLI (connectorctl / Glue)	Covers the core operational verbs; usable by technical operators
License / control server	Key issuance, pilot grants, Stripe billing, portal auth all functional
REST API	Core agent, session, memory, and decision endpoints working
Structured logging + Prometheus metrics	Operational observability baseline in place

Table 9.1: Technology components that are production-grade by end of 2026

9.1.2 What is genuinely immature by end of 2026

Gap	Why it matters commercially
Multi-node / cluster support	Enterprises with more than one team or environment cannot standardise on Connector without this
Enterprise auth integrations (SSO, SAML, SCIM)	Healthcare and fintech procurement will block on the absence of these
Compliance evidence tooling (SOC2, HIPAA artefacts)	Buyers need machine-generated compliance evidence, not just logs
CLS compiler maturity	CLS is architecturally defined but the compiler pipeline is immature; governed workflow authoring is still hard
Entitlement model consistency	Internal model has fragmentation between licensing server, platform server, and billing service; this creates product confusion
Developer onboarding friction	Time-to-first-value for a new technical user is still too high
Distributed tracing (OpenTelemetry end-to-end)	Tracing works but is not yet an out-of-the-box enterprise integration
UI completeness	Dashboard, portal, and admin are functional but unpolished

Table 9.2: Genuine technology gaps at the end of 2026

9.2 2027: Infrastructure Grade

9.2.1 The theme

2027's technology theme is **infrastructure grade**: every feature that pilots accepted despite friction must become reliable, enterprise-integrable, and fast to onboard. The standard to aim for is that a new enterprise customer should go from contract signature to first governed workflow in under 30 days with no custom engineering from the Connector team.

9.2.2 Priority technology investments in 2027

P1 — Enterprise auth: SSO (SAML 2.0, OIDC), SCIM user provisioning, team-level RBAC

P1 — Compliance tooling: SOC2 evidence generation, HIPAA BAA support, audit log export

P2 — Multi-node alpha: cluster deployment, node-to-node trust, distributed agent scheduling

P2 — CLS compiler v2: stable contract syntax, governance pre/post conditions, YAML authoring surface

P3 — Entitlement model unification: single canonical model across all services, no fragmentation

P3 — Onboarding improvement: guided setup, glue doctor workflow, time-to-value under 30 min

P4 — OpenTelemetry integration: end-to-end distributed tracing exportable to Datadog, Grafana

P4 — UI polish: dashboard, portal, admin — production-quality, consistent design system

Figure 9.1: 2027 technology investment priorities — P1 through P4

9.2.3 Technology architecture in 2027

The fundamental architecture should not change in 2027. The Memory Kernel, predicate chain, and two-server design remain the core. What changes is the **integration surface** around the core: enterprise systems expect to find SAML endpoints, SCIM APIs, audit log webhooks, and structured compliance evidence that existing security and compliance toolchains can consume.

Architectural principle for 2027

Do not redesign the core. Add the perimeter. The memory kernel and policy chain are the moat. The perimeter is what enterprise procurement checks before they sign.

9.2.4 2027 technology SWOT

Dimension	Honest 2027 assessment
Strengths	Rust runtime is genuinely fast and memory-safe; content-addressed memory kernel is architecturally ahead of log-based competitors; governance-in-runtime is a real differentiator; single-node deployment is simple and auditable
Weaknesses	Rust limits the developer talent pool significantly; multi-node complexity is non-trivial to implement correctly; CLS is still a research-grade concept in a production-grade product; UI tech (Leptos/WASM) is unfamiliar to most frontend engineers
Opportunities	New engineering hires post-Seed 1 can be directed at the highest-value gaps; enterprise customers often become co-designers of compliance tooling (free R&D); Rust ecosystem for AI/ML tooling is growing
Threats	Engineering velocity may suffer if multi-node is underestimated; a technically similar but better-funded competitor could execute the same architecture faster with more engineers

Table 9.3: Technology SWOT for 2027

9.2.5 Honest technical risks in 2027

Technical risk register for 2027

- **Multi-node is not just a feature.** Distributed systems introduce consensus, split-brain, and partial-failure modes. If the team underestimates this, multi-node will be shipped half-baked and will generate enterprise incidents that damage reputation. Fix: do an alpha with 2 enterprise accounts before general availability.
- **CLS authoring experience.** If governed workflow authoring requires deep Rust knowledge, Connector will be unusable by the people who actually write business logic. Fix: the YAML / domain-language surface for CLS must be the default path by end of 2027.
- **Entitlement fragmentation creates bugs.** Three different tier models in different parts of the codebase will eventually produce a customer-visible billing or access bug. Fix: entitlement unification is not optional; it is P3 but must not slip past mid-2027.

9.3 2028: Platform Maturity and Architecture Expansion

9.3.1 The theme

2028's technology theme is **platform maturity**: the runtime becomes something that enterprise platform teams can build *on top of*, not just use as a point solution. This requires stable APIs, SDK-grade developer surfaces, meaningful extensibility, and the beginning of an ecosystem.

9.3.2 Priority technology investments in 2028

Investment area	What it enables	Priority
Multi-node GA	Multi-team, multi-environment deployments; enterprise rollout	Critical
SDK surface (Python, TypeScript)	AI engineers author agents and policies in familiar languages	Critical
CLS contracts in production	Governed workflows with pre/post conditions, evidence trails, auditability	High
Agent marketplace / distribution	Internal enterprise agent libraries; partner integrations	Medium
Advanced compliance modules	GDPR, EU AI Act evidence generation; automated compliance reports	High
API versioning and stability guarantee	Enterprise integrations require API stability commitments	High
Cell-to-cell protocol (CNP cross-node)	Multi-cell agent collaboration; distributed governed workflows	Medium
Model-agnostic runtime	Connector governs workflows regardless of whether model is GPT, Claude, Gemini, Llama	High

Table 9.4: 2028 technology investment priorities

9.3.3 The SDK question

The 2026–2027 developer experience is CLI-first and Rust-first. By 2028, this is a commercial liability. The teams writing AI agents are using Python and TypeScript. If Connector requires a Rust developer to author governance policies or integrate the runtime, it will be excluded from a very large part of the market.

The solution is not to rewrite the runtime in Python. The solution is to build a **thin, idiomatic client SDK** in Python and TypeScript that wraps the existing REST API and Glue surface, with full type coverage and real developer experience investment.

SDK surface target for 2028

- **connector-py**: Python SDK — agent creation, session management, memory writes, decision recording, audit retrieval.
- **connector-ts**: TypeScript SDK — same surface, designed for Node.js and browser-adjacent environments.
- Both SDKs must support the full Glue lifecycle grammar, not just raw API calls.

9.3.4 2028 technology SWOT

Dimension	Honest 2028 assessment
Strengths	Stable multi-node runtime; SDK surface makes Connector accessible to Python/TypeScript teams; CLS contracts provide a differentiated governance authoring experience; signed receipt chain is auditor-ready
Weaknesses	API surface is large and will require ongoing maintenance commitment; multi-node adds operational complexity for customers running self-hosted; cell-to-cell protocol is still early; compliance module scope creep risk
Opportunities	Enterprise platform teams adopting Connector as the “runtime of record” can generate large expansion ARR; open-source community edition could dramatically expand top-of-funnel; regulatory compliance deadlines create urgency in new geographies
Threats	OpenAI, Anthropic, or Google may add governance and audit features to their enterprise APIs, reducing the need for a separate runtime; Kubernetes-native observability tools may expand into agent governance territory

Table 9.5: Technology SWOT for 2028

9.3.5 Honest technical risks in 2028

Technical risk register for 2028

- **API compatibility debt.** Once enterprise customers depend on the REST API at scale, breaking changes become costly. The team must commit to a versioning and deprecation policy in 2027 before this becomes an active problem.
- **Operational complexity growth.** Multi-node, multi-cell, and multi-model support each add dimensions of failure. Without a strong internal testing and reliability programme, the platform will accumulate subtle distributed failures that damage enterprise trust.
- **Model-agnosticism is harder than it sounds.** Different model providers have different token formats, context window behaviours, and error modes. Truly model-agnostic governance requires an abstraction layer that is currently not in the architecture.
- **SDK quality.** A poorly-written SDK is worse than no SDK. If the Python or TypeScript SDK is a thin wrapper with bad ergonomics, it will generate negative developer reputation. Fix: invest engineering time or hire a developer experience specialist before launching.

9.4 2029 and Beyond: Architecture Stability and Ecosystem

9.4.1 The theme

By 2029, the core architecture should be stable and well-understood enough that the focus shifts from *building the runtime* to *building the ecosystem around the runtime*. This mirrors what happened with Kubernetes (2016–2020), Stripe (2013–2017), and Snowflake (2015–2019): the platform stabilises, the ecosystem grows, and the platform company captures

more value through partnerships, integrations, and vertical depth than through continued core feature development.

9.4.2 Technology evolution by 2029

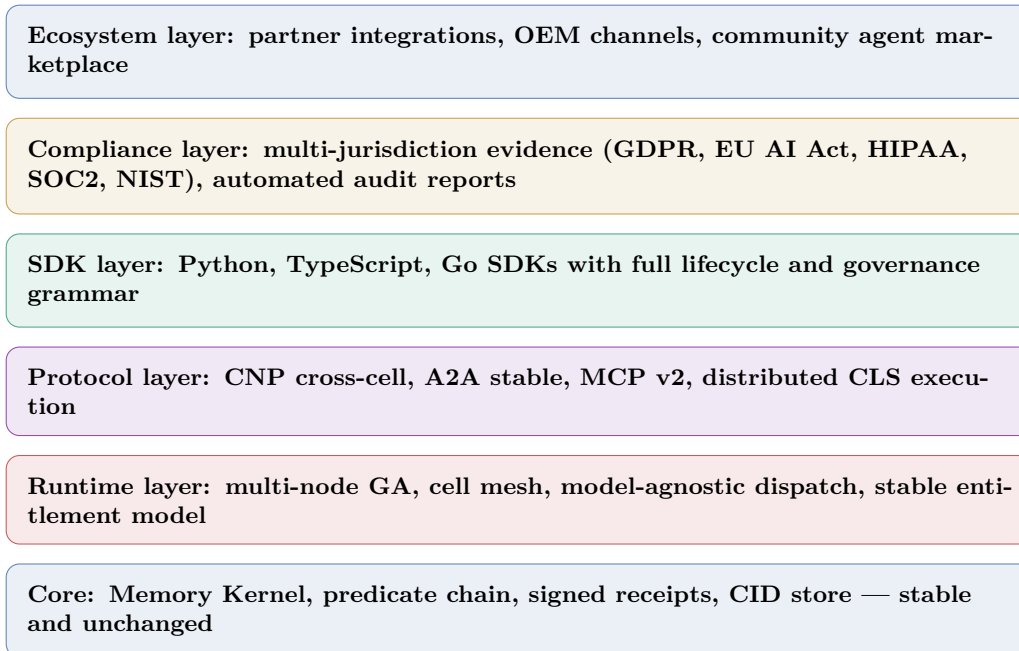


Figure 9.2: Connector technology stack by 2029 — core stable, ecosystem above it

9.4.3 Open-source strategy question

By 2029, Connector will face a strategic decision that most successful infrastructure companies have faced: whether to open-source part of the platform to accelerate adoption. The historical evidence is instructive:

Company	Open-source model	Outcome
HashiCorp	Open-source core (Terraform, Vault), commercial enterprise	Dominant category, \$6.4B acquisition by IBM in 2024
Elastic	Open-source (Elasticsearch), commercial cloud	\$1B+ ARR; later relicensed to defend against AWS
Grafana Labs	Open-source (Grafana), commercial cloud + enterprise	\$6B valuation; massive developer community
Kong	Open-source API gateway, commercial enterprise	\$1.4B valuation; deep enterprise penetration

Table 9.6: Open-source infrastructure company models — historical reference

For Connector, the **recommended path** is an open-core model: open-source the Memory Kernel and base runtime under a permissive or BSL licence, and keep multi-node clustering, enterprise compliance tooling, the full CLS compiler, and the commercial control server as proprietary. This matches the HashiCorp and Grafana models.

9.4.4 2029+ technology SWOT

Dimension	Honest 2029 assessment
Strengths	Stable, well-documented architecture; multi-year enterprise deployments create deep switching costs; open-core community drives top-of-funnel; compliance tooling is years ahead of competitors who start later; Rust performance and safety guarantees are a genuine enterprise selling point
Weaknesses	Open-source creates commoditisation risk for the core; large team means higher coordination overhead; architectural complexity (11-layer predicate chain, CLS compiler, multi-node) makes onboarding new engineers slow; technical debt from early shipping decisions is accumulating
Opportunities	AI governance standards body involvement; OEM partnerships with EHR and HRMS vendors; international expansion enabled by multi-jurisdiction compliance modules; developer community creates enterprise pipeline without marketing spend
Threats	Cloud providers (AWS, Azure, GCP) may build managed agent governance services and leverage distribution moat; EU AI Act could be superseded or weakened, reducing regulatory urgency; the market could consolidate around 2–3 vendors before Connector reaches category leadership

Table 9.7: Technology SWOT for 2029+ stabilisation and ecosystem stage

9.4.5 Honest technical risks in 2029

Technical risk register for 2029+

- **Architectural ossification.** Once the runtime is stable and enterprise customers depend on it, the team will resist the changes needed to support new AI interaction models (e.g. long-horizon planning, new context protocols). Fix: maintain a research/architecture function separate from product maintenance.
- **Commoditisation from below.** A well-executed open-source project from a well-funded startup or foundation could replicate the memory kernel and governance model. Fix: the proprietary value must be in the compliance tooling, enterprise integrations, and the commercial control plane — not just the core runtime.
- **Team scale problems.** 30–40 engineers on one codebase creates coordination problems. The architecture needs clear ownership boundaries (memory kernel team, protocol team, SDK team, compliance team) or velocity will degrade.
- **AI capability shifts.** The AI landscape in 2029 may look very different from 2026. Models with dramatically better self-governance or built-in auditability could reduce the need for an external runtime layer. Fix: monitor the frontier; build governance tooling that *integrates with* model-level evidence rather than competing with it.

9.5 Technology Evolution Summary

Year	Core focus	Must-ship	Biggest risk
2026	Single-node production-grade	Pilot tooling, stable node boot, basic compliance	Pilot quality; product too raw for enterprise
2027	Infrastructure grade	SSO/SAML, multi-node alpha, CLS v2, entitlement unification	Multi-node underestimated; compliance gaps block sales
2028	Platform maturity	Multi-node GA, Python/TS SDKs, API stability, model-agnostic runtime	SDK quality; API versioning debt; operational complexity
2029+	Ecosystem and stability	Open-core decision, partner integrations, compliance modules	Commoditisation; architectural ossification; AI capability shifts

Table 9.8: Technology evolution summary by year

9.6 The Compound Relationship Between Business and Technology

The chapters above treat business evolution and technology evolution as separate narratives. In practice they are not. Every commercial milestone depends on a specific technology readiness, and every technology investment should be justified by a specific commercial unlock.

Commercial goal	Technology prerequisite	If technology is not ready
Close first enterprise annual contract (2027)	SSO and SOC2 evidence	Deal stalls in procurement or fails security review
Expand first customer from one workflow to business unit (2027)	Multi-node alpha + team-level RBAC	Expansion is technically blocked
Close Series A or Seed 2 (2028)	\$1M+ ARR + SDK-grade DX + demonstrable multi-node	ARR is limited by single-node constraint; investor confidence is lower
OEM or partnership channel (2029)	Stable versioned API + Python SDK + compliance module APIs	Partners cannot build on an unstable surface
Series B or strategic acquisition (2030)	Mature ecosystem, enterprise reference base, open-core community	Without ecosystem, the company is a feature, not a platform

Table 9.9: Commercial goals and their direct technology prerequisites

The one sentence that ties business and technology together

Every year that Connector ships the right technology *slightly ahead* of when a customer needs it, the commercial motion accelerates. Every year it ships *behind*, it leaves deals on the table to better-resourced competitors. The difference is planning, not engineering talent.

Chapter 10

Execution Reality and the Five-Year Journey

10.1 The Gap Between a Good Plan and a Real Business

The previous chapters document what Connector should build and how the commercial motion should unfold. This chapter documents the part that most business plans skip: **what actually makes execution hard**, where most companies fail in practice, and what it takes in concrete monthly operational terms to cross \$1M ARR by the end of 2028.

The data in this chapter is grounded in published benchmarks from OpenView’s 2024 SaaS Benchmarks report, SaaS Capital’s B2B cohort studies, Bessemer Venture Partners’ State of the Cloud 2025 report, and Andreessen Horowitz’s enterprise GTM guides. None of this is theoretical. Every number here reflects observed outcomes from comparable companies.

The central honesty this chapter requires

Most founders can build the product. Very few can sell it, retain customers, and grow ARR on a lean team simultaneously. The failure mode for infrastructure and developer-tools companies is almost never “the product was bad.” It is “the product was good but no one bought it fast enough before the team ran out of runway.”

10.2 The Real Difficulty of Enterprise B2B Sales

10.2.1 Why enterprise sales is harder than expected

A deal with a regulated enterprise — healthcare, fintech, or enterprise AI platform — is not a single decision. It is a **multi-stakeholder bureaucratic process** that typically involves:

- A **technical champion** who evaluated the product and wants it.
- An **economic buyer** (usually VP or C-level) who controls the budget.
- A **security and compliance reviewer** who will ask for SOC2, pen test results, and data processing agreements.
- A **legal team** who will redline the contract and SLA.
- A **procurement team** who will run a vendor review process.
- **IT or platform engineering** who must approve the deployment model.

All six must say yes. Any one can say no. In healthcare, this process averages **9–18 months** from first conversation to signed contract. In fintech it averages **6–12 months**. In enterprise platform teams it averages **4–9 months**.

Vertical	Avg. sales cycle	Main blockers	Key relationships needed
Healthcare	9–18 months	HIPAA BAA, security review, procurement, legal	CISO, CTO, VP Engineering, Compliance
Fintech / insurtech	6–12 months	SOC2, audit controls, vendor risk, contract terms	CTO, Risk Officer, Head of AI
Enterprise platform teams	4–9 months	Internal security review, budget approval, pilot scope	Head of AI, VP Platform, Engineering Director

Table 10.1: Realistic enterprise sales cycle benchmarks by vertical

10.2.2 What this means for cash and runway

If Connector closes its first 5 enterprise annual contracts over a 12-month period starting in Q1 2027, the ARR impact of those deals does not appear until Q4 2027 or Q1 2028. This means the company carries cost (salaries, tooling, compliance spend) for the entire duration before seeing the revenue. Typical burn during this period is \$80k–\$120k per month with a 6–8 person team.

At a \$2M Seed 1 raise:

- At \$80k/month burn: **25 months of runway** (tight but workable)
- At \$120k/month burn: **17 months of runway** (dangerously short)

The practical implication: **do not hire ahead of revenue**. Every hire before \$500k ARR is a bet that the deal pipeline will materialise on schedule. Historically, it does not. OpenView’s 2024 cohort data shows that B2B SaaS companies with fewer than 10 employees take an average of 27 months to reach \$1M ARR. Companies that over-hire early run out of runway at month 18–20.

10.3 The Arithmetic of \$1M ARR by End of 2028

10.3.1 The math in plain terms

\$1M ARR is not one deal. It is a portfolio of deals that must be acquired, retained, and expanded over approximately 24 months from a standing start. Here is the concrete arithmetic:

ACV assumption	Customers needed for \$1M ARR	Pipeline needed (3:1 close ratio)	Conversations needed (10:1 to pipeline)
\$15k ACV	67 customers	200 deals in pipeline	2,000 conversations
\$25k ACV	40 customers	120 deals in pipeline	1,200 conversations
\$35k ACV	29 customers	87 deals in pipeline	870 conversations
\$50k ACV	20 customers	60 deals in pipeline	600 conversations
\$80k ACV	13 customers	39 deals in pipeline	390 conversations

Table 10.2: What \$1M ARR requires at different ACV levels

The clear implication: **pricing discipline is an operational requirement, not a pricing decision**. A company selling at \$15k ACV needs to run a volume motion that a 6-person team physically cannot sustain. The realistic path to \$1M ARR by end of 2028 is 25–35 customers paying \$28k–\$40k ACV.

10.3.2 Month-by-month ARR build plan

Period	Target ARR	Milestones required	re-	Key risk
End of Q4 2026	\$80k–\$150k	3–5 annual conversions from pilots		Pilot-to-annual conversion rate under 30%
End of Q2 2027	\$200k–\$300k	6–10 paying customers, Seed 1 closed		Seed 1 delayed; cash gap
End of Q4 2027	\$400k–\$600k	15–20 paying customers, repeatable motion starting		First sales hire still not closing independently
End of Q2 2028	\$650k–\$800k	20–27 customers, expansion ARR from 2027 cohort		NRR drops below 100% from first cohort churn
End of Q4 2028	\$900k–\$1.2M	28–35 customers, 2 verticals proven, Seed 2 or Series A in process		Sales cycle longer than modelled; miss the gate

Table 10.3: Quarterly ARR milestones on the path to \$1M by end of 2028

10.3.3 The expansion ARR multiplier

The difference between a company that reaches \$1M ARR in 2028 and one that gets stuck at \$500k is almost always **net revenue retention**. If customers expand from a single workflow at \$20k ARR to a business-unit deployment at \$60k ARR, the ARR grows without new customer acquisition cost.

The land-and-expand motion for Connector specifically:

- **Land:** One workflow, one team, \$20–30k ARR pilot-converted annual.
- **Expand step 1:** Add second workflow or second team, \$45–60k ARR.
- **Expand step 2:** Business-unit-wide standardisation, \$80–120k ARR.
- **Platform:** Company-wide mandate with custom SLA, \$150k+ ARR.

If just 30% of the first 15 customers expand from land to step 1 by end of 2028, that adds \$90k–\$135k ARR with zero new customer acquisition cost. This is the difference between reaching \$900k and reaching \$1.1M.

10.4 The Five Most Dangerous Execution Failure Modes

These are not hypothetical risks. They are the documented causes of failure in comparable B2B infrastructure companies, ranked by frequency in post-mortems collected by First Round Capital, YC, and Sequoia from 2019–2024.

10.4.1 Failure mode 1: ICP sprawl

What happens: The team closes a healthcare deal, a fintech deal, an e-commerce deal, and a defence contractor deal across the first 18 months. Every deal requires a different message, different integrations, different compliance posture, and different support knowledge. The team cannot build expertise in any one vertical. Reference customers cannot speak to each other. The company cannot build a playbook.

The data: Companies that focus on two or fewer verticals in years 1–2 reach \$1M ARR 40% faster than those that spread across four or more, according to OpenView’s 2023 expansion stage cohort.

The fix: Enforce the ICP with budget. Only run pilots in healthcare, fintech, and enterprise platform teams. Decline adjacent opportunities regardless of how attractive they look. The exception is a deal \$100k+ ACV that directly funds the next quarter — but that is not a pivot, it is a one-off.

10.4.2 Failure mode 2: Feature building without sales validation

What happens: Engineering builds the features customers asked for in discovery calls. Three months later, those features do not close deals because the customer’s buying criteria changed or the champion left. The product has grown but the revenue has not.

The data: In Bessemer’s 2024 DevTools company study, 67% of features built in response to customer requests before \$500k ARR were rated “not important” to the buying decision by the same customers who requested them.

The fix: Only build a feature if it unlocks a specific deal that is in late stage right now. Every other roadmap item is deferred until that deal is signed. The question before any engineering sprint is: “Which deal does this close?”

10.4.3 Failure mode 3: Underpricing to close

What happens: The team drops price to close a stalled deal. The customer pays \$10k instead of \$25k. Six months later, the team has ten customers at \$10k ARR and \$100k ARR total — not enough to raise, not enough to sustain, not enough to hire. The pricing signal also tells the market that Connector is a cheap tool, not an enterprise governance layer.

The data: Paddle's 2024 SaaS pricing study found that companies that discount more than 20% in the first year take an average of 11 additional months to reach \$1M ARR compared to companies that hold price and accept a lower close rate.

The fix: Hold list price. Offer extended payment terms or a longer pilot as an alternative to discounting. A customer who needs a 50% discount to buy is not a reference customer — they are a support burden at a price that does not sustain the business.

10.4.4 Failure mode 4: The wrong first sales hire

What happens: The founder, exhausted by selling, hires an enterprise AE from a larger company. The AE expects inbound leads, a marketing team, a polished pitch deck, and existing category awareness. None of these exist. The AE cannot generate their own pipeline. After six months and \$120k in salary, the hire is reversed. The company has lost six months and \$120k.

The data: First Round Capital's 2023 sales hiring study found that 75% of enterprise AE hires at sub-\$500k ARR companies fail within nine months. The correct first sales hire at this stage is **not** an enterprise AE — it is an outbound-first, technically capable seller who can run their own pipeline and also do implementation support.

The fix: The correct profile for Connector's first sales hire is: 3–5 years at a developer tools or infrastructure startup, comfortable with technical buyers, capable of running 20–30 outbound conversations per week independently, willing to do a 30-day paid trial with quota from month one.

10.4.5 Failure mode 5: Losing the first customer cohort

What happens: The first 5–8 enterprise customers were founder-sold with high-touch support. When the founder moves on to new deals and an engineer takes over support, the customer experience degrades. The champion leaves. The contract is not renewed. The company has 0% NRR and must replace the lost ARR with new customers just to stay flat.

The data: SaaS Capital's 2024 cohort study found that companies with NRR below 90% in year 2 have a less than 15% probability of reaching \$3M ARR without a fundamental product or GTM change.

The fix: The first CS hire or onboarding engineer is not optional. It should happen at the same time as or before the first sales hire. The moment a customer signs, they need a named contact, a success plan, and a quarterly check-in. This is not a nice-to-have — it is the ARR retention infrastructure.

10.5 How to Make Connector a Real Product (Not a Demo)

The gap between a working pilot and a real product is not a feature gap. It is an **operational readiness gap**. Real products have:

Requirement	What it means for Connector	Target year
Time-to-value under 30 min	A new developer can install the node, create an agent, and see a governed decision in under 30 minutes	2027 Q1
Self-service onboarding	No call required to start a pilot; documentation and <code>glue doctor</code> are sufficient	2027 Q1
Upgrade path	v1.x to v2.x with a documented migration guide and backward-compatible API	2027 Q3
SLA-backed uptime	99.5% uptime commitment for hosted; deployment scripts that meet this for self-hosted	2027 Q2
Status page	Public status page showing current and historical uptime	2027 Q2
Support tiers	Email SLA (Team), dedicated Slack (Enterprise), emergency paging (Enterprise Plus)	2027 Q2
Security documentation	Trust page: SOC2 Type I certificate, pen test summary, data flow diagram	2027 Q2
Changelog	Public changelog updated with every release	2026 Q4
Pricing page	Clear, public pricing with no hidden fees	2026 Q4

Table 10.4: Operational readiness checklist — what converts a demo into a product

None of these are engineering challenges. They are **discipline challenges**. Most founders deprioritise them in favour of features. The customers who churn most often in year 2 are the ones who experienced a product that felt like a demo, not an infrastructure investment.

10.6 The Five-Year Journey: One Honest Arc

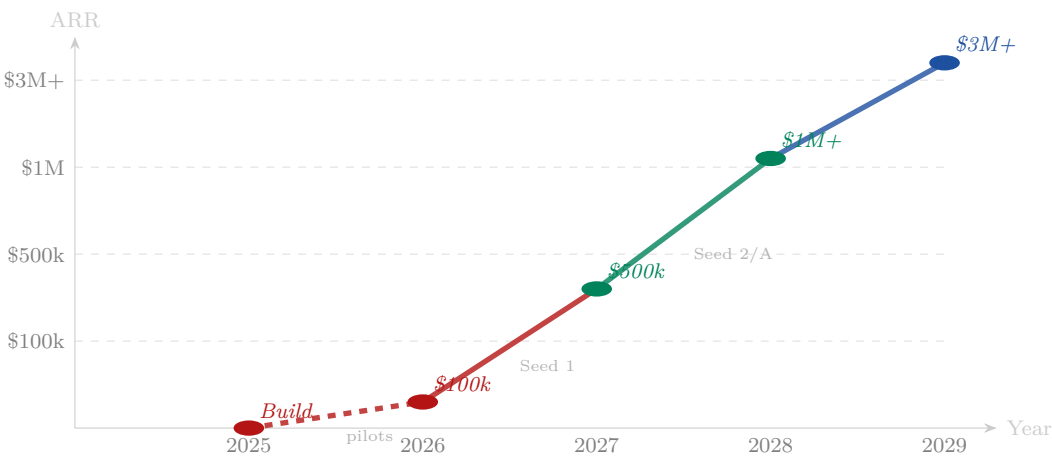


Figure 10.1: Realistic ARR trajectory 2025–2029 — base case

10.6.1 Year 1 (2025): Foundation and pre-product-market fit

Reality: The team is building. There is no revenue. Every decision is a technical bet on what the market will want. The single most important output of this year is *not code* — it is the identification of one workflow in one industry where an identifiable buyer is in pain and the current solutions are inadequate.

What success looks like: A working single-node deployment with a governing runtime, a CLI that a skilled engineer can use without documentation, and at least three engineers or AI practitioners who have tried it and said it solves a real problem they have today.

Honest failure mode: Building all seven layers of the architecture to production quality before talking to a single paying customer.

10.6.2 Year 2 (2026): Wedge validation — the most critical year

Reality: This is the year where the company either finds the wedge or does not. Ten paid pilots sounds achievable in a plan. In practice, getting ten organisations in regulated industries to pay \$15k+ for software from a company with no brand, no case studies, and no analyst coverage requires **50–80 founder-led conversations**, a compelling demo that works in a live environment, and a pricing structure that passes procurement scrutiny.

What success looks like:

- 8–10 paid pilots signed in 2026 across two verticals.
- At least 3–5 of those converting to annual contracts.
- A reference customer willing to speak at a conference or be named in collateral.
- A clear answer to: “*What makes a pilot succeed vs. stall?*”

Honest failure mode: Running 8 pilots but closing none to annual because the pilot success criteria were unclear and the champion had no internal budget authority.

10.6.3 Year 3 (2027): From wedge to repeatable — the hardest transition

Reality: Year 3 is where most good technical companies break. The founder has been selling. Seed 1 has been raised. There is now pressure to “scale.” But the product still requires founder involvement to close deals, the ICP is not crisp enough for a new hire to execute on, and the first engineering hires are now senior enough to want to build things the founder is not yet ready to fund.

What success looks like:

- A sales hire who closes their first 3 deals without founder support.
- A documented ICP with a specific job title, industry, company size, and technical environment that a new team member can act on in week one.
- ARR crossing \$500k with at least 15 distinct paying customers.
- NRR above 110% — the first cohort is expanding, not churning.

Honest failure mode: The Seed 1 hire class (engineers + sales + CS) burns the raise in 14 months without \$500k ARR. The company must raise a down-round bridge or shut down.

10.6.4 Year 4 (2028): The \$1M ARR gate — the fundraising proof point

Reality: \$1M ARR is not a vanity milestone. It is the minimum bar for a credible Series A conversation with most institutional investors in 2026–2028. Below \$1M ARR, the narrative is “we believe this will work.” Above \$1M ARR with 100%+ NRR and two proven verticals, the narrative is “this is working.” The narrative difference translates to \$10M–\$15M valuation difference.

What success looks like:

- 28–35 paying customers by Q4 2028.
- Average ACV at \$28k–\$40k, held with pricing discipline.
- NRR at 110–125%, driven by workflow expansion within existing accounts.
- Series A or Seed 2 process initiated in Q3 2028 with a data room showing 18 months of ARR growth history.
- Product GA on multi-node; SOC2 Type II in progress.

Honest failure mode: The company is at \$750k ARR in Q4 2028 with good product and happy customers but no path to \$1M before the runway ends. This is a fundraising and execution problem, not a product problem. Fix: raise a \$1.5M Seed 2 bridge in Q2 2028, hire one more outbound seller, and extend runway to Q3 2029.

10.6.5 Year 5 (2029): Platform entry — from product to platform

Reality: By 2029, Connector is either a recognised category player or a strong point solution that has not yet broken through to category definition. The distinction matters because category players raise capital at 15–20x ARR while point solutions raise at 6–8x ARR. The gap in funding round size determines whether Connector can hire the engineering team needed to build the ecosystem layer.

What success looks like:

- \$3M–\$5M ARR with 120%+ NRR and 50–80 customers.
- Inbound deal flow that supplements outbound — the category is starting to find Connector, not just Connector finding the category.
- 2–3 vertical depth case studies published and referenced in analyst coverage.
- An open-core decision made with board alignment.
- Series A or B closed, providing 24+ months of runway.

Honest failure mode: The market consolidates around a competitor with more funding before Connector reaches category definition. This is the scenario that Series A capital is supposed to prevent.

10.7 The Overall Five-Year Reality: What the Numbers Show

Year	ARR target	Team	Capital state	Primary risk
2025	\$0	1–3 founders	Pre-raise; boot-strapped or angel	Building the wrong product
2026	\$80k–\$150k	2–4 people	Pre-seed or angel funded	Cannot close pilots at price
2027	\$400k–\$600k	6–10 people	Seed 1 raised (\$1.5M–\$3.5M)	First hire class fails; burn too high
2028	\$900k–\$1.2M	12–20 people	Seed 2 or Series A process	Sales cycle longer than modelled
2029	\$3M–\$5M	25–40 people	Series A or B raised	Category competitor with more funding

Table 10.5: Five-year reality summary — ARR, team, capital, and primary risk at each stage

10.8 SWOT of Execution Itself

This is separate from the product SWOT and technology SWOT in earlier chapters. This is the SWOT of **the company’s ability to execute** the plan described above.

Dimension	Honest execution assessment
Strengths	Technical founders who have built the hardest parts of the product already; regulated industry focus gives natural pricing power; early pilot program creates real customer relationships before formal sales; the two-server architecture is a natural enterprise-trust story
Weaknesses	No brand, no analyst coverage, no inbound pipeline; founder-led sales is not scalable past 20 customers; enterprise trust in a two-year-old company takes time; compliance costs (\$50k–\$150k for SOC2) are material at pre-revenue stage; Rust reduces hiring pool
Opportunities	Regulatory pressure (EU AI Act, US NIST AI RMF) creates urgency that shortens enterprise sales cycles; AI governance is still poorly served — buyers are actively searching; first-mover reference customers in healthcare or fintech become defensible moat if retained
Threats	Running out of runway before reaching \$500k ARR is the single most likely terminal risk; a well-funded competitor enters the same wedge with 10x the GTM budget; the AI market shifts and agent governance becomes a free feature from hyperscalers

Table 10.6: Execution SWOT — separate from product and technology SWOTs

10.9 The One Sentence That Defines Whether This Works

All of the analysis in this document — the architecture, the GTM motion, the pricing model, the pilot design, the technology roadmap — reduces to a single operational truth:

The single sentence that determines whether Connector succeeds

Connector succeeds if and only if a founder-led team of fewer than ten people can close 25–35 enterprise customers at \$28k–\$40k ACV by end of 2028, retain them at above 110% NRR, and use that proof to raise the capital needed to scale past what a small team can sustain.

Everything else is detail.

The product is ready to support this. The architecture is right. The market timing is real. The question is entirely one of **execution discipline**: holding the ICP, holding the price, not over-hiring, not building features that do not close deals, and keeping the first customer cohort healthy until the team is large enough to scale.

That is what the next three years require.

Chapter 11

Complete Platform Architecture Reference

11.1 Full-Stack Component Map

All sixteen runtime engines, the VAC Memory Kernel, CLS compiler pipeline, Glue surface, CNP/A2A/MCP protocol stack, control-plane services, and storage infrastructure are shown together in the diagram below. Colour coding: blue = kernel core, green = execution engines, purple = CLS and logic, orange = protocols, teal = developer surfaces, amber = control plane, grey = infrastructure. Data flow moves top-to-bottom. Layer numbers are on the right margin.

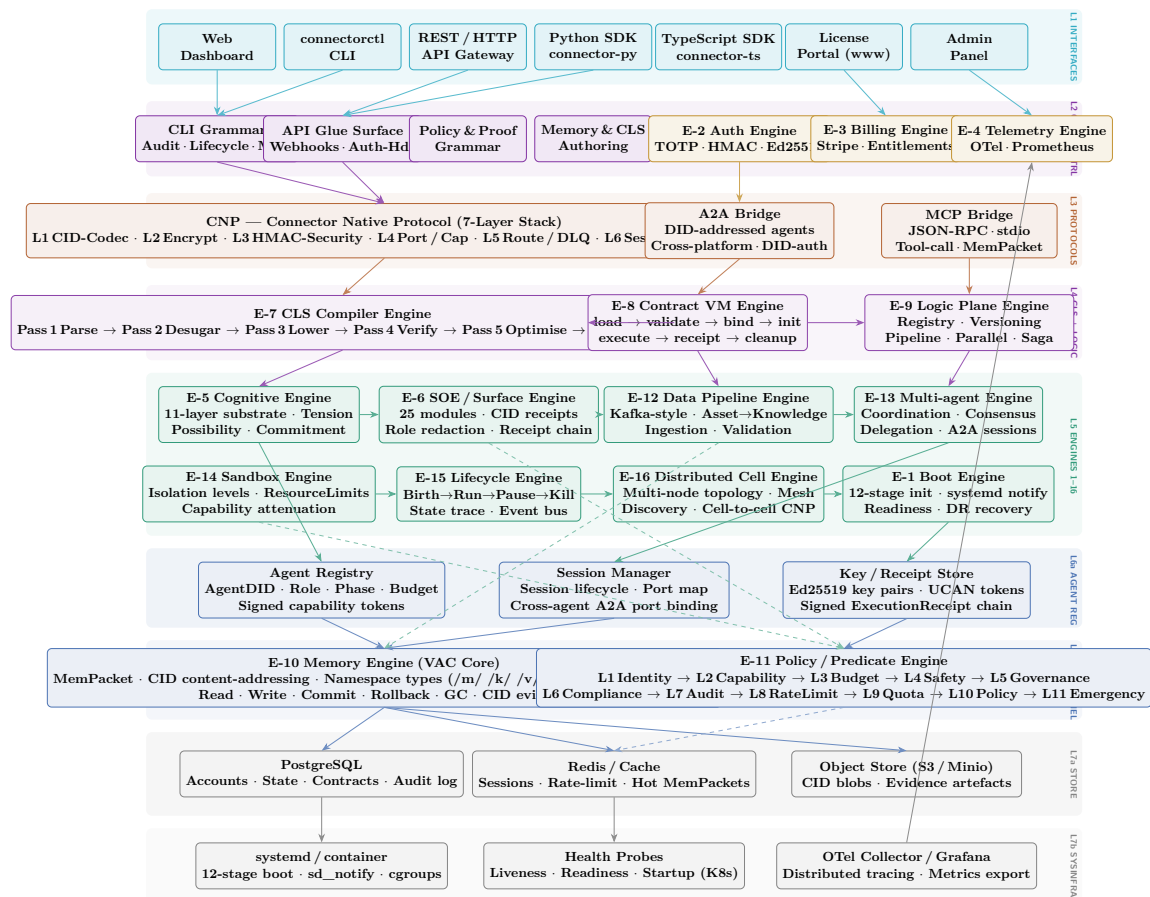


Figure 11.1: Complete Connector platform architecture — all 16 engines, VAC Memory Kernel, CLS compiler, Glue surface, CNP / A2A / MCP protocol stack, and storage substrate

11.2 The Sixteen Runtime Engines

#	Engine	Responsibility	Layer
E-1	Boot Engine	12-stage node initialisation, systemd sd_notify, DR recovery	L5
E-2	Auth Engine	TOTP, HMAC, Ed25519 key verification, session tokens	L2
E-3	Billing Engine	Stripe Checkout, webhook entitlement updates, tier enforcement	L2
E-4	Telemetry Engine	OpenTelemetry, Prometheus metrics, usage tracking, audit sink	L2
E-5	Cognitive Engine	11-layer substrate: tension, possibility, commitment, plan, reflection	L5
E-6	SOE / Surface Engine	25-module surface system: CID receipts, role redaction, receipt chain	L5
E-7	CLS Compiler Engine	6-pass compilation: Parse→Desugar→Lower→Verify→Optimise→Emit	L4
E-8	Contract VM Engine	Solution Contract execution lifecycle, signed receipt production	L4
E-9	Logic Plane Engine	ContractRegistry, Pipeline / Parallel / Saga / MapReduce composition	L4
E-10	Memory Engine	MemPacket, CID content-addressing, namespace types, GC	L6
E-11	Policy / Predicate Engine	11-layer enforcement chain evaluated per kernel operation	L6
E-12	Data Pipeline Engine	Kafka-style asset→knowledge ingestion, validation, tagging	L5
E-13	Multi-agent Engine	Coordination patterns, consensus, delegation, A2A sessions	L5
E-14	Sandbox Engine	Isolation levels, resource limits, capability attenuation per agent	L5
E-15	Lifecycle Engine	Birth→Run→Pause→Kill state machine with full event trace	L5
E-16	Distributed Cell Engine	Multi-node topology, mesh, discovery, cell-to-cell CNP routing	L5

Table 11.1: The sixteen Connector runtime engines with layer assignment

11.3 Key Data-Flow Paths

Path	Component sequence
Agent decision	CLI / API → Glue → CNP → CLS Compiler → Contract VM → Cognitive Engine → Memory Engine → Policy Engine → Signed Receipt → Storage
Tool invocation	Agent → MCP Bridge → Logic Plane → Sandbox Engine → Tool call → MemPacket → Memory Engine
Cross-agent message	Agent A (CNP port) → CNP L4 capability check → CNP L5 route → Agent B port → Session Manager → Receipt
Cross-platform A2A	Agent (CNP) → A2A Bridge (DID-auth) → Remote agent → Response → MemPacket logged
Data ingestion	Asset (/v/) → Pipeline Engine → Validation → Knowledge (/k/) → Memory Engine → CID committed
Boot sequence	systemd → Boot Engine (12 stages) → Policy Engine load → Agent Registry init → Session Manager → READY

Table 11.2: Key data-flow paths through the platform